

Introduction to Hypersonic Aerodynamics (Chapter 12)

Objectives:

Understand characteristics of hypersonic flow

Newtonian and Modified Newtonian flow – uses and limitations

Understand flow characteristics of stagnation region

Evaluate lift, drag, and pitching moment for simple geometries

Introduction to heating and transition

Examples of Hypersonic Vehicles

X-15 hypersonic test aircraft exceeded $M = 5$ in flight

1961 – Gagarin $M = 25$ in reentry ($\sim$ as the Shuttle)

1969 – Apollo $M = 36$ in reentry (~ 45 reentry from Mars)

1971 SR-71 $M=3.6$

2004 X-43A $M=9.6$

2010 X-51 scramjet demonstrator ($M=6$)

$M > 5$ just a rule of thumb

hypersonic flow: high Mach number regime where certain physical flow phenomena become progressively more important as the Mach # is increased

nothing magic happens when M goes from 4.99 $\rightarrow$ 5.01

Some features:

High Mach numbers $M_{\infty} = \frac{U_{\infty}}{a_{\infty}} \gg 1$ (12.1)

In wind tunnels low T and a *(speed of sound = $\sqrt{\gamma R T}$)*

Downstream of a normal shock wave:

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_1^2}{(\gamma - 1)M_1^2 + 2} \quad (8.75)$$

For high Mach numbers $\Rightarrow \varepsilon = \frac{\rho_1}{\rho_2} = \frac{\gamma - 1}{\gamma + 1}$ (12.3)

Value of γ : *for high T $\Rightarrow \gamma \rightarrow 1 \Rightarrow \varepsilon \rightarrow 0$*

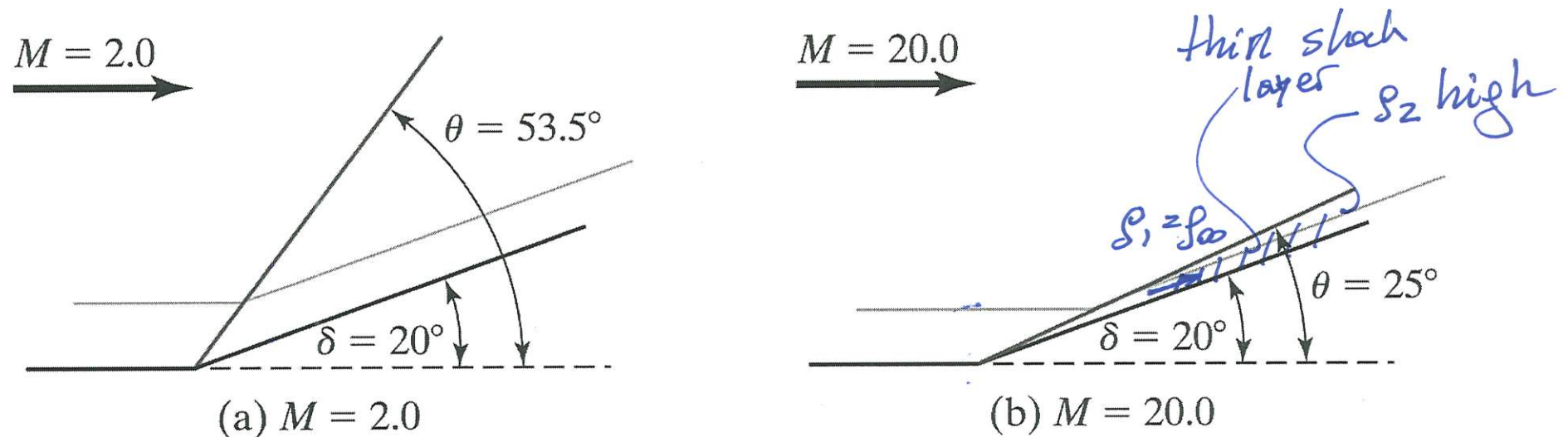
For slender configurations a strong shock when:

$$M_{\infty} \sin \theta_{\infty} \gg 1$$

Distinguishing Characteristics [12.2]

Thin Shock Layers:

Shock closer to the surface



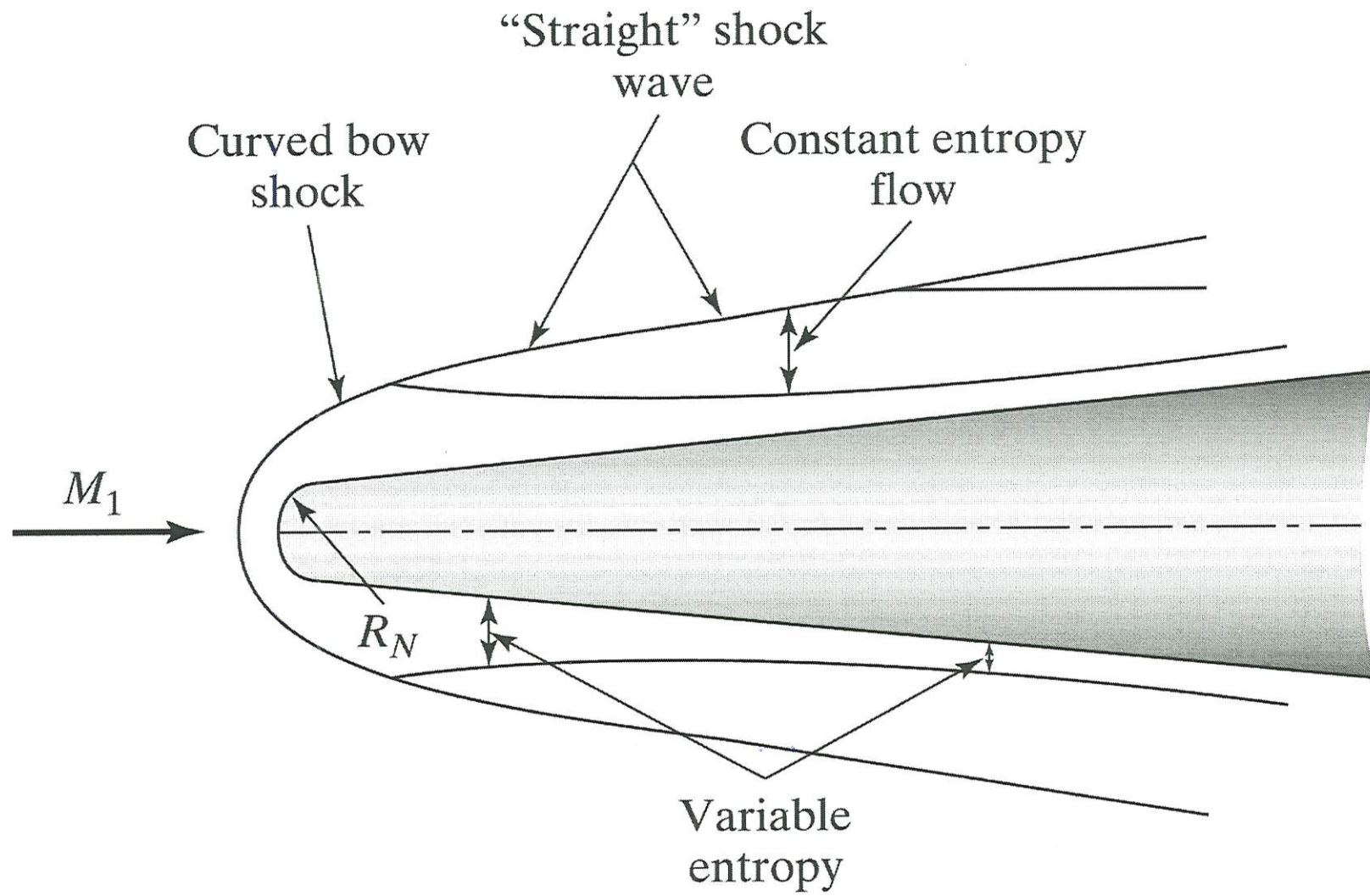
Thicker boundary layers: $\delta \propto \frac{M^2}{\sqrt{\text{Re}}}$

Shock layer: flow field between shock wave & body surface $\Rightarrow$ thin
for hypersonic speed (boundary layer can be thick $\Rightarrow$ merging with
shock layer $\Rightarrow$ fully viscous layer)

Entropy layers:

Entropy varies if shocks are not straight

On a blunt-nosed body shock wave is slightly detached highly
curved shock $\Rightarrow$ flow behind shock not isentropic; entropy layer
varying strength $\leadsto$ different ΔS
interacts with boundary layer and increases heating



Viscous-Inviscid Interactions

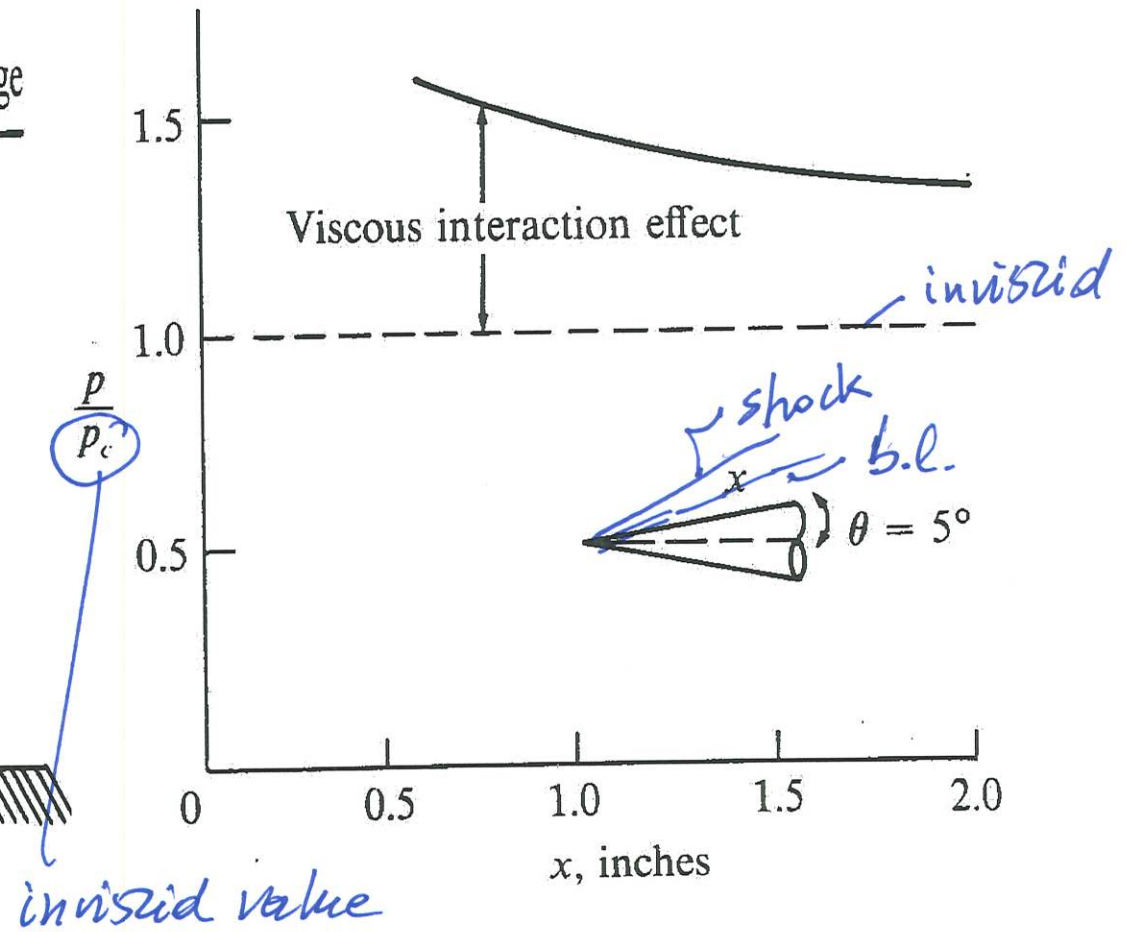
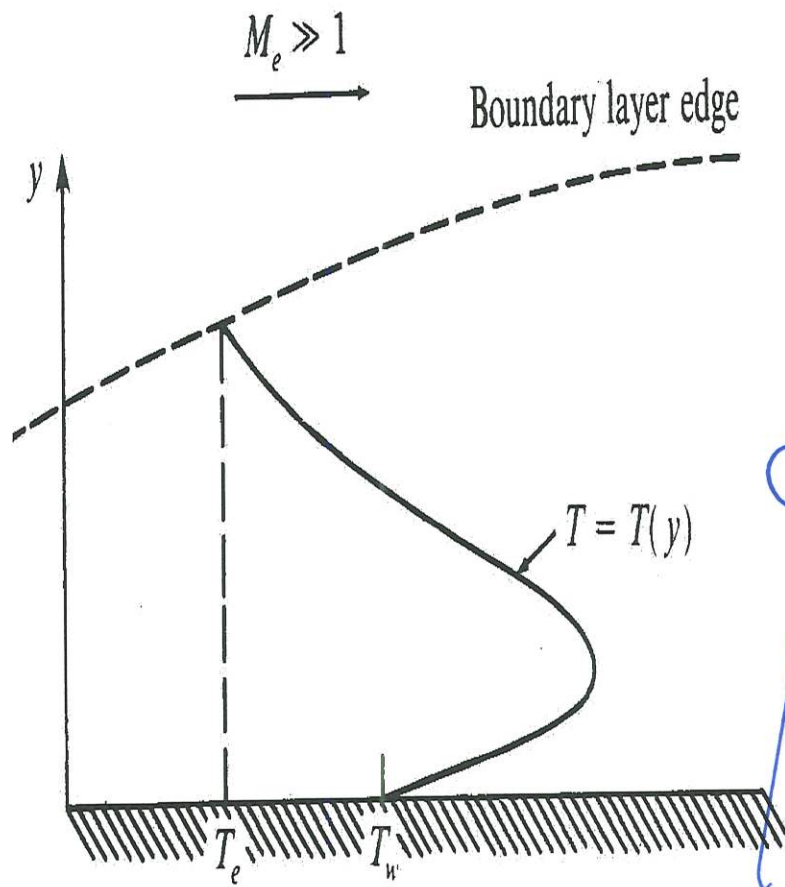
Boundary layer thickness: $\delta \propto \frac{M^2}{\sqrt{Re_x}}$

at very high Mach numbers δ can become very large *compared to $\delta_{incompressible}$*
 viscous interaction: thick boundary layer flow affects outer inviscid
 flow (changes in inviscid flow influence boundary layer)

Example of flow over a cone below.

Also, a factor in Columbia accident.

compromised
 (foam hit the thermal
 protection system $\rightarrow$ hot
 gas went inside)



High Temperature Effects

Ratio of kinetic energy to gas internal energy: (per unit mass of gas)

$$\frac{U^2 / 2}{u_e} = \frac{U^2 / 2}{c_v T} = \frac{\gamma - 1}{2} \frac{U^2}{RT} = \frac{\gamma(\gamma - 1)}{2} \frac{U^2}{a^2} = \frac{\gamma(\gamma - 1)}{2} M^2$$

$$c_v = \frac{R}{\gamma - 1}$$

$$a = \sqrt{\gamma R T}$$

when $M_\infty \gg 1$ the kinetic energy is $\gg$ internal energy.

In a boundary layer, kinetic energy becomes internal energy

gas temperature $\propto$ internal energy $\Rightarrow$ rapid increase of T

Example: Apollo re-entry:

$M_\infty = 36$, altitude = 75Km ($\sim 250,000$ ft), $T_\infty = -10^\circ\text{C} = 263^\circ\text{K}$

From normal shock tables $\Rightarrow \frac{p_2}{p_1} = 1512 \quad \frac{T_2}{T_1} = 252.9 \quad \Rightarrow T_2 = 252.9 \times 263 = 66,513^\circ\text{K}$

The result is *not realistic*

Non-constant γ

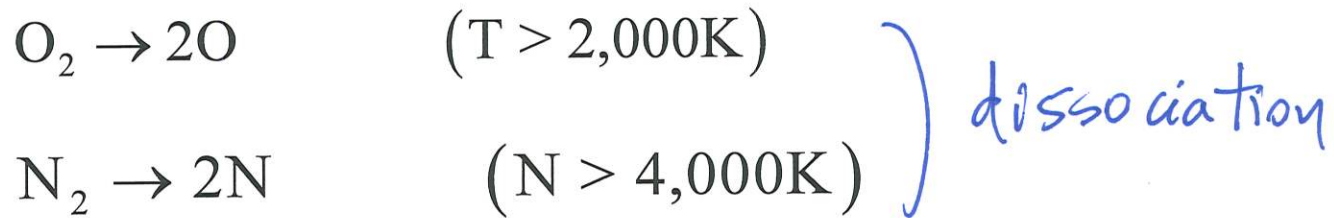
At 300°K $\gamma = 1.4$, at $2,500^\circ\text{K}$ $\gamma = 1.2$ high T $\gamma \rightarrow 1$

chemical reactions/dissociation

radiative heat transfer

High-temperatures produce chemical reactions:

Molecular vibrations ($T > 800 \text{ K}$)



At higher temperature ($T > 9,000 \text{ K}$) atoms will ionize:



plasma

These phenomena happen during reentry.

Variations in the flow-field around blunt-nosed bodies

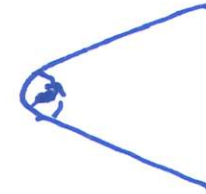
($\gamma \neq 1.4$ all tables not valid) increase of radiative heating

Heating $\propto \frac{1}{\sqrt{R_{l.e.}}} \Rightarrow$ For low heating we need a blunt body

$R_{l.e.}$ = leading edge
radius



low heat



high heat
(small $R_{l.e.}$)

For low heating = need a blunt body

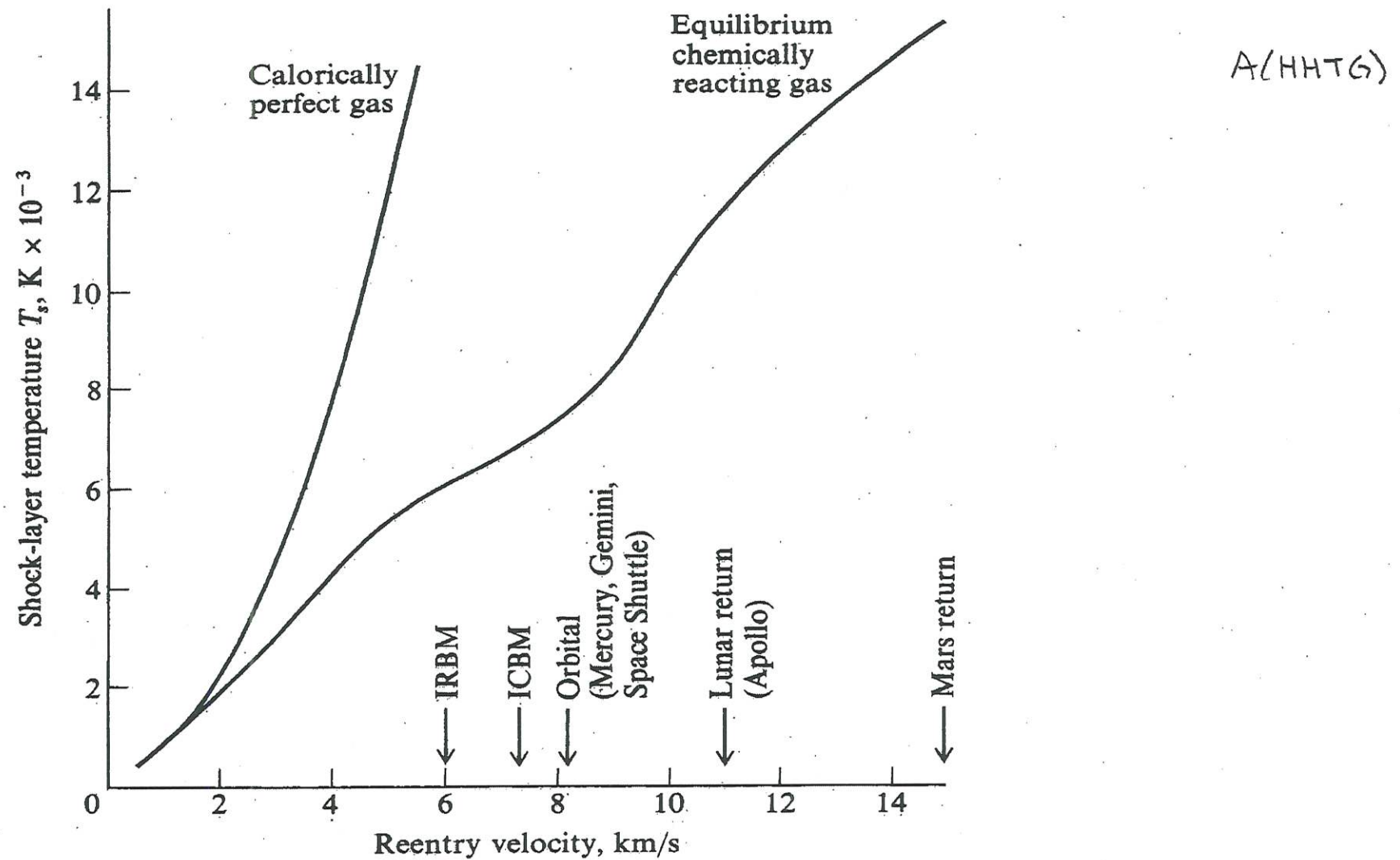
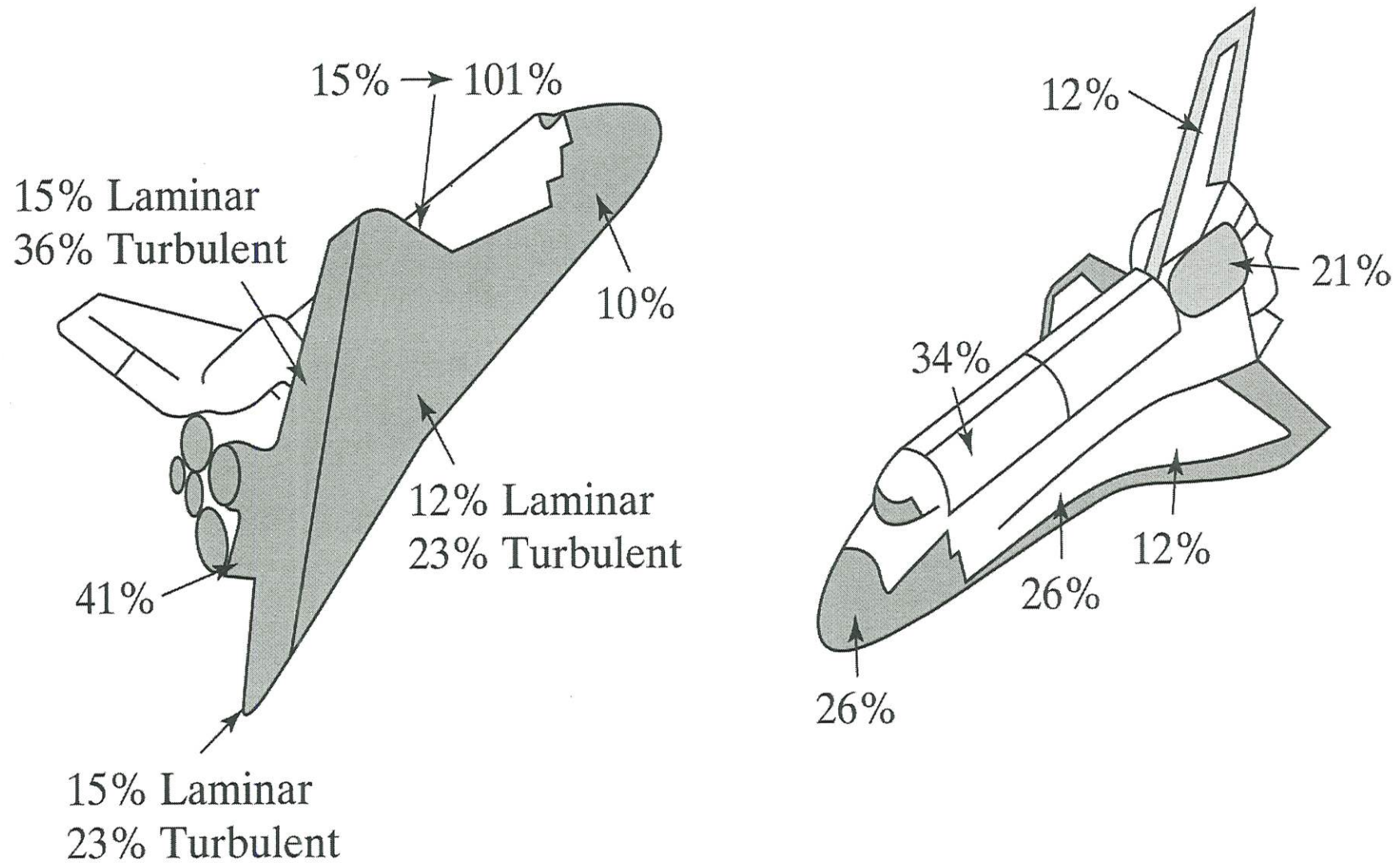


FIGURE 1.18

Temperature behind a normal shock wave as a function of free-stream velocity at a standard altitude of 52 km. (From Ref. 4.)



Uncertainties of heating estimates for the Shuttle

Low density flows

At high altitudes (e.g. 300,000 ft) the air is not continuous

mean free path λ : average distance between collisions

for standard conditions $\lambda = 2.176 \times 10^{-7}$ ft (small!) $\Rightarrow$

“continuum” assumption

At 342,000 ft $\lambda = 1$ ft $\Rightarrow$ low density flow

Knudsen number $Kn \equiv \frac{\lambda}{\ell} \leftarrow (\text{characteristic flow dimension})$

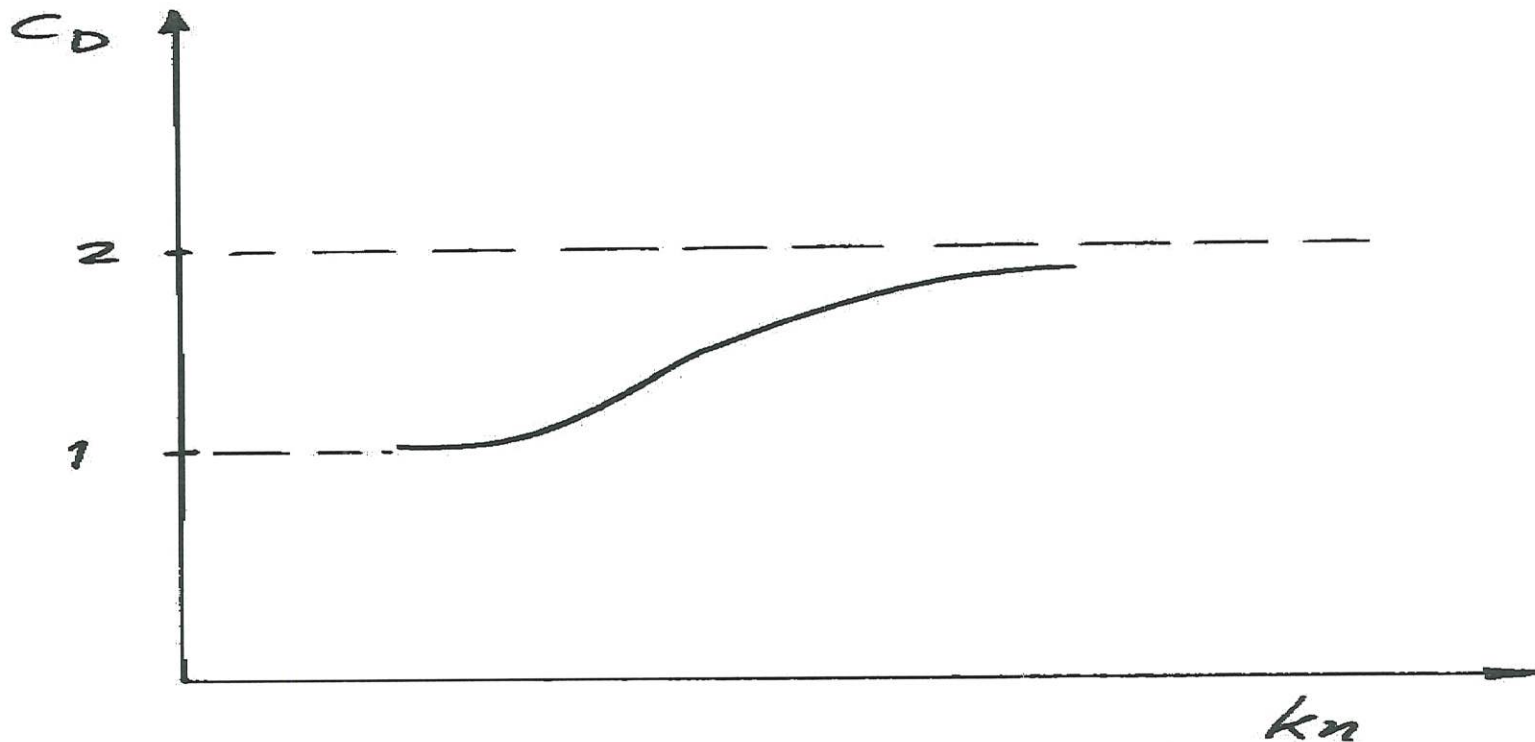
for $Kn \leq 0.03 \Rightarrow$ continuum (Navier Stokes eqs - no slip condition)

$0.03 < Kn < 0.2$ slip conditions on the boundary

$Kn > 10$ free molecule flow

Boltzmann equation; Direct Simulation Monte Carlo (DSMC)

Forces and moments f (kn), e.g. C_D of sphere



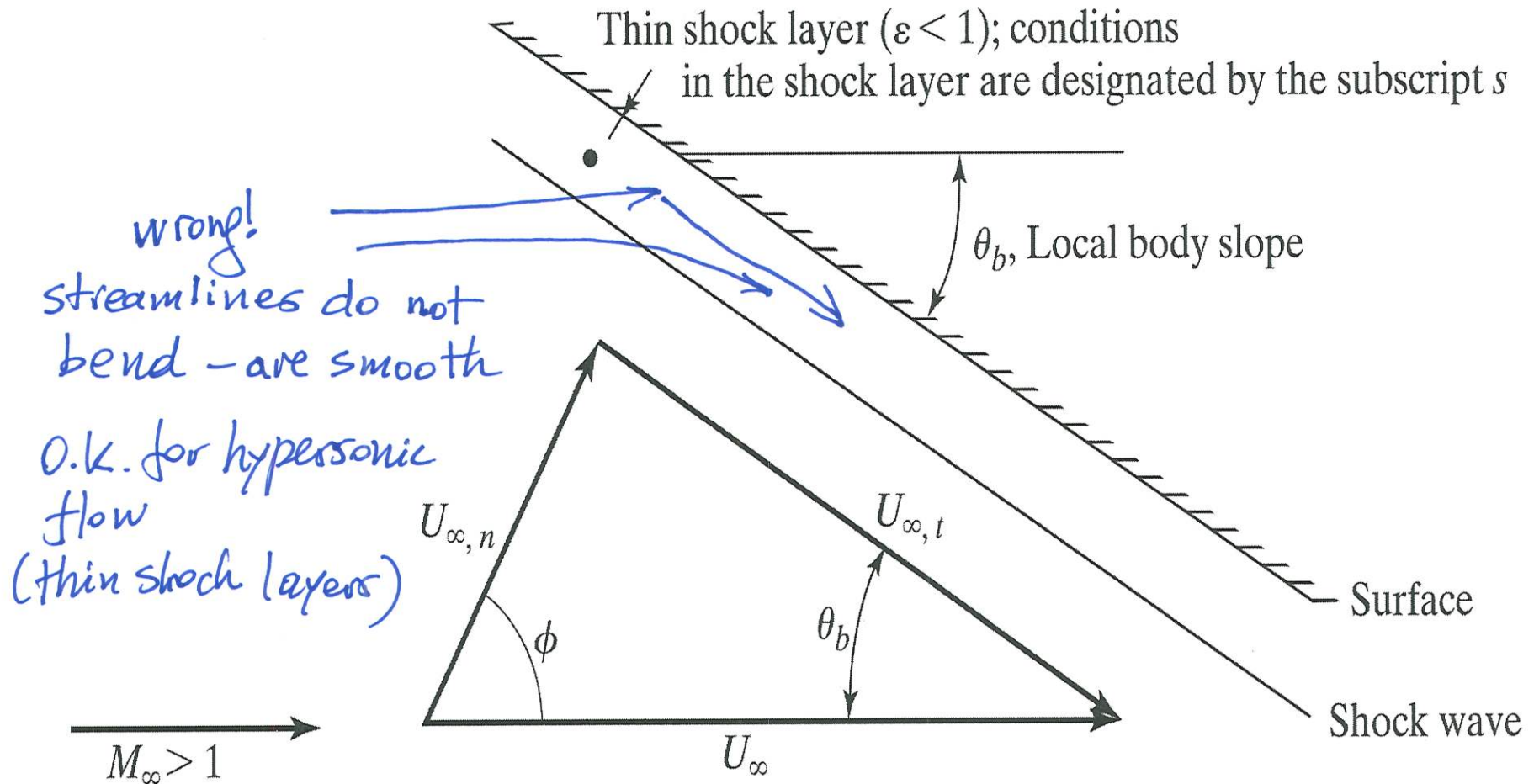
Most of important reentry at low altitudes

Some flight in low density

Effects also important for small characteristic dimensions-MEMS

Newton's law for Hypersonic Flow [8.2]

1687 Newton's idea of a fluid:



transfer of normal momentum to the surface (tangential momentum is preserved). Conservation of normal momentum:

$$p_{\infty} + \rho_{\infty} U_{\infty}^2 \sin^2 \theta = p_s \quad (12.5)$$

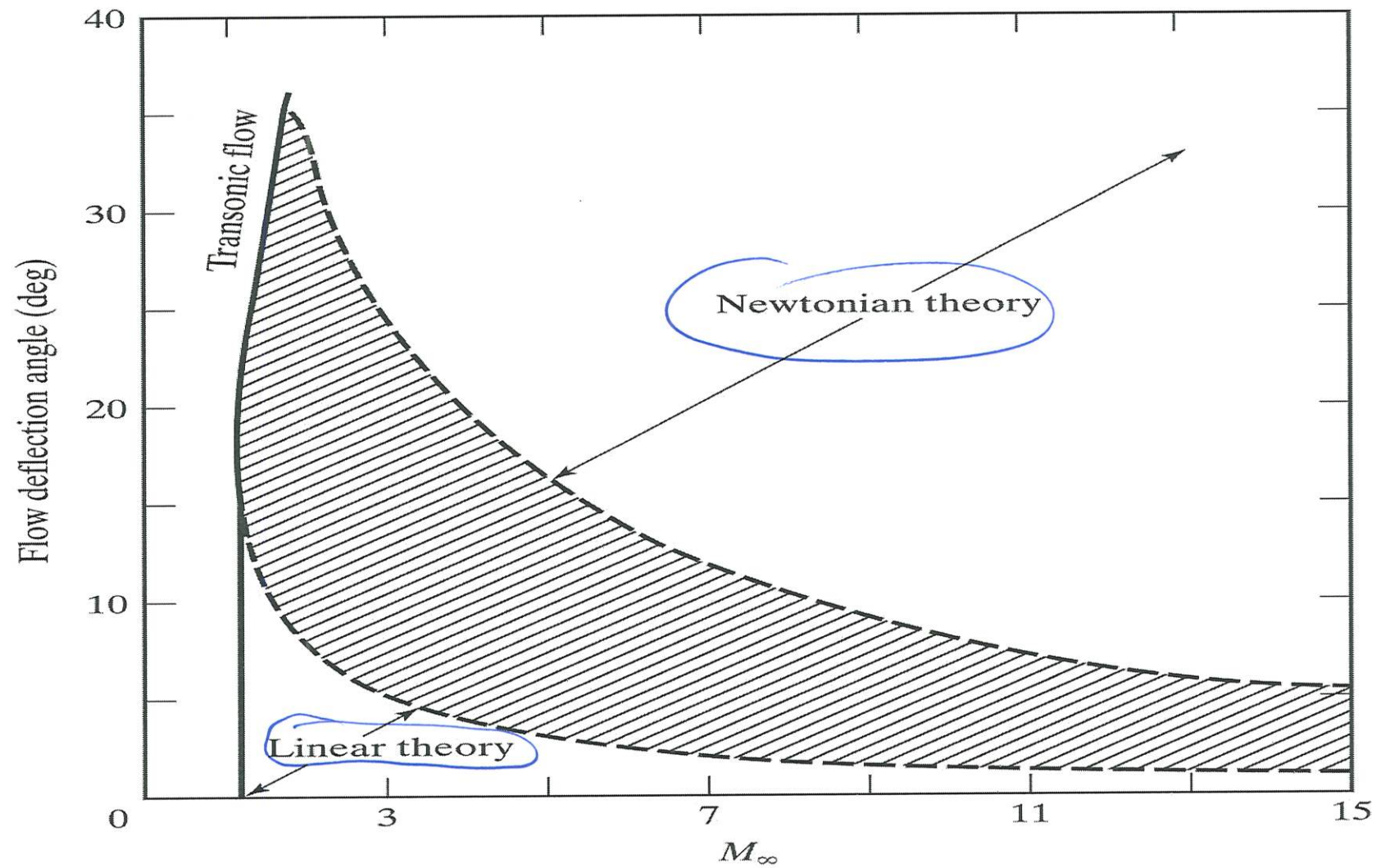
$$\Rightarrow C_p = \frac{p_s - p_{\infty}}{\frac{1}{2} \rho_{\infty} U_{\infty}^2} = 2 \sin^2 \theta_b = 2 \cos \phi \quad (12.6)$$

This equation was derived to find resistance of bodies (e.g. ships).

Small shock layer; from (12.3) $\gamma \rightarrow 1$

Also: $M_{\infty} \sin \theta_{\infty} \gg 1$

$$\epsilon = \frac{p_1}{p_2} = \frac{\gamma - 1}{\gamma + 1} \rightarrow 0$$



Applicability of Newtonian theory

high M, θ

$C_p = 0$ (vacuum) for “shadow” surfaces

Newton's theory predicts $C_{p_{\max}} = 2$ (for incompressible $C_p = 1$)

1-D normal shock:

shock

$p_\infty = p_1$	p_2
$\rho_\infty = \rho_1$	ρ_2
$U_\infty = U_1$	U_2

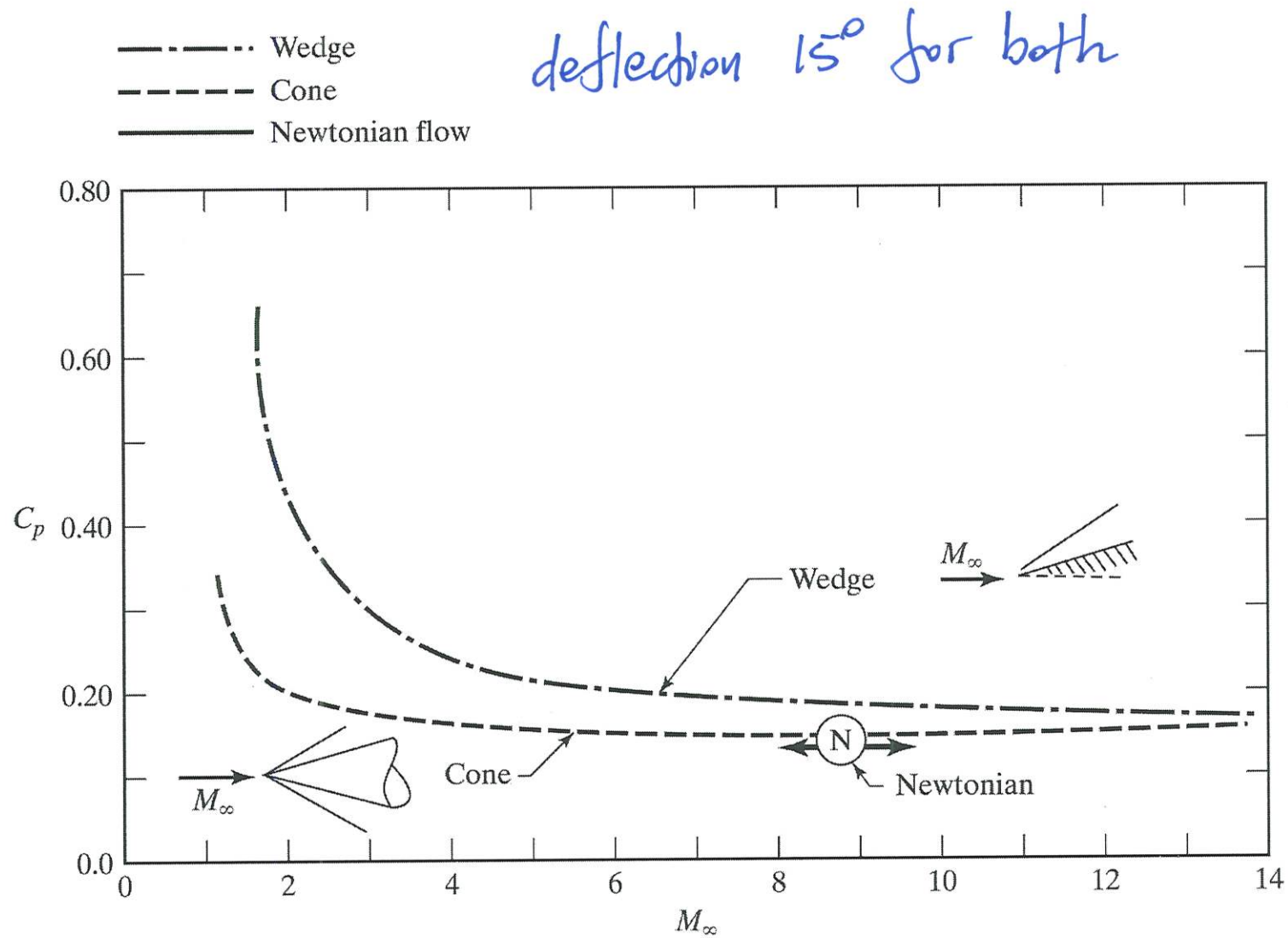
from B's eq.

$$\frac{p - p_\infty}{\frac{1}{2} \rho V_\infty^2}$$

Momentum equation:

$$p_\infty + \rho_\infty U_\infty^2 = p_2 + \rho_2 U_2^2 \approx 0 \quad (M_\infty \gg 1)$$

$$\Rightarrow C_p = \frac{p_2 - p_\infty}{\frac{1}{2} \rho_\infty U_\infty^2} = 2$$



At high Mach numbers: Mach number independence

Lift and Drag of Flat Plates

Incompressible:

$$C_\ell = 2\pi(\alpha - \alpha_{Lo})$$

$$C_D = C_d + \frac{C_\ell^2}{\pi e AR}$$

Supersonic:

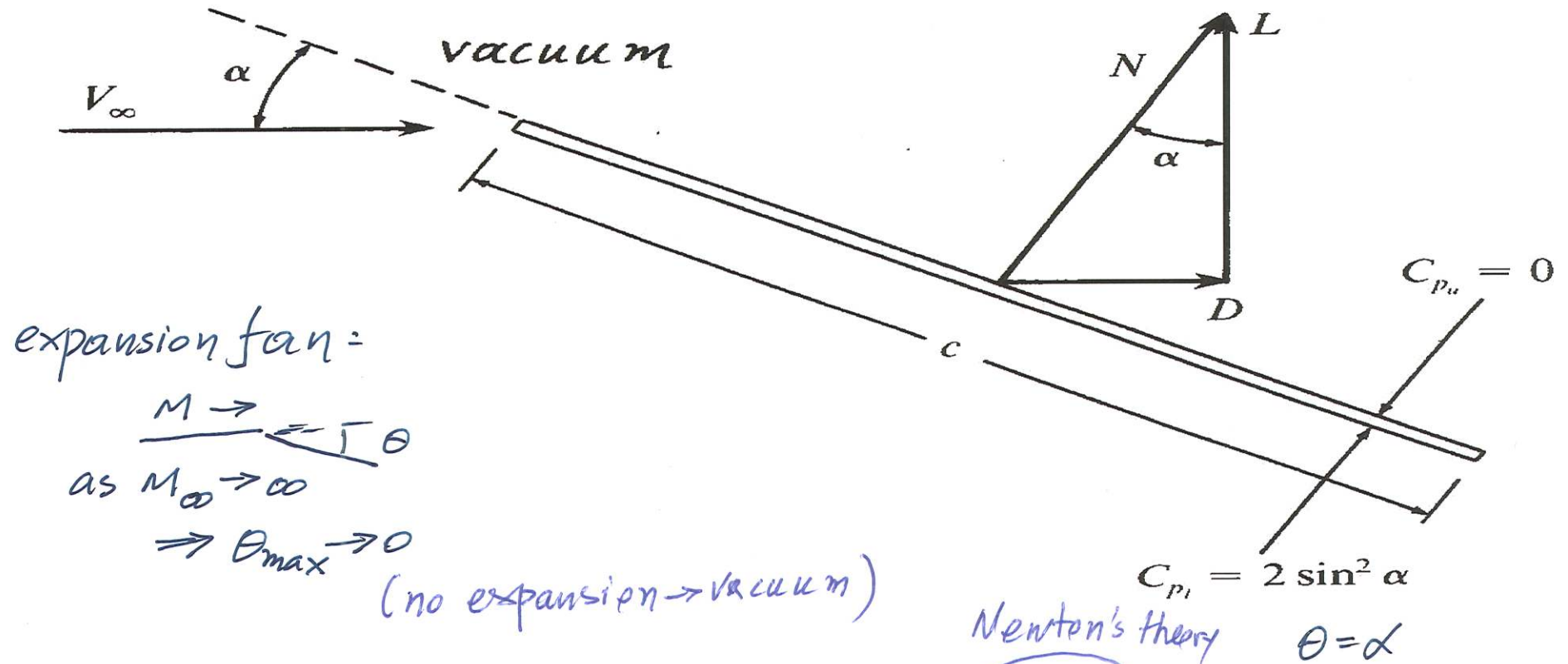
$$C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

independant of shape

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}}$$

shock-wave drag

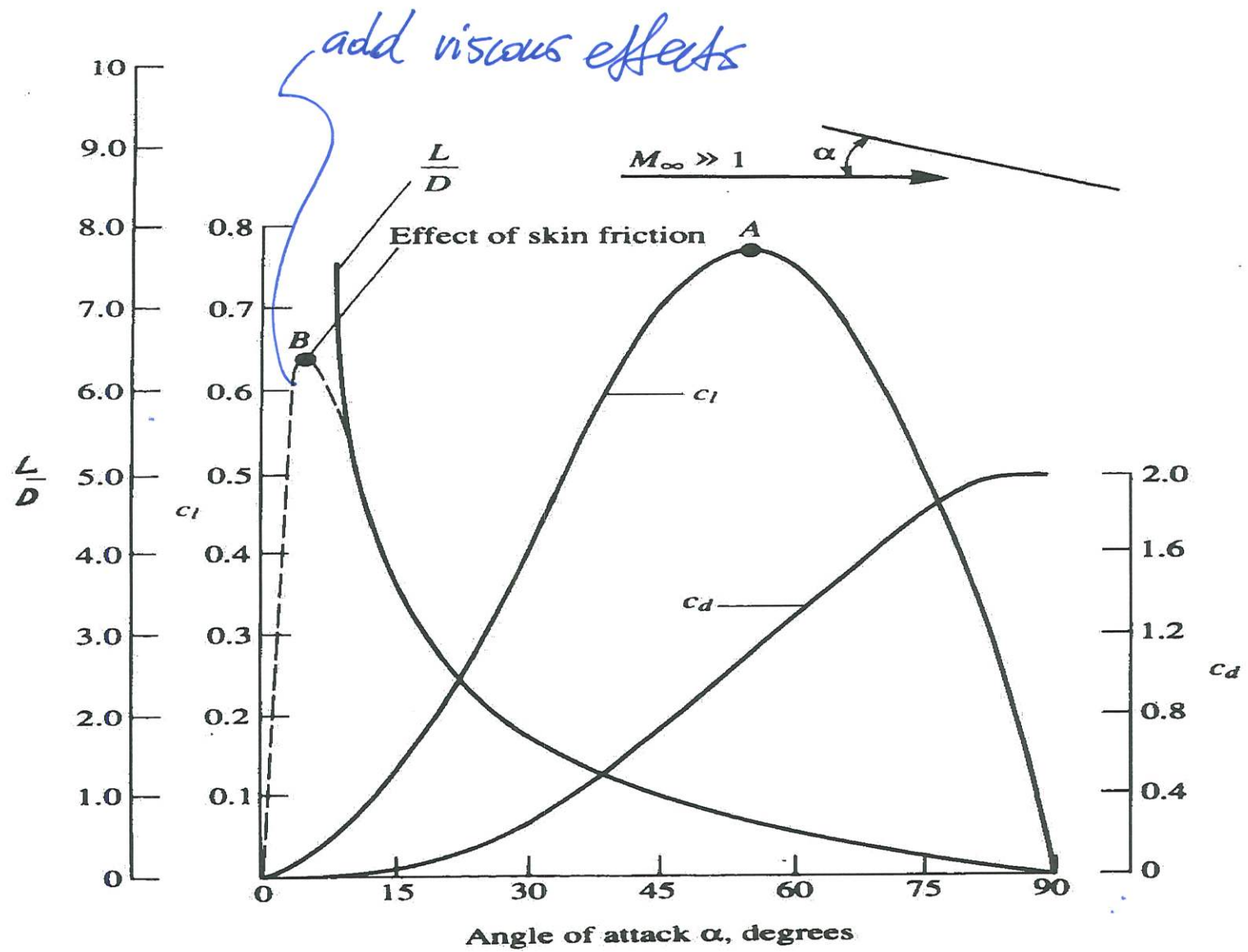
Hypersonic -Newtonian theory



$$L = N \cos \alpha = \rho_\infty V_\infty^2 A \sin^2 \alpha \cos \alpha \Rightarrow C_\ell = 2 \sin^2 \alpha \cos \alpha$$

$$D = N \sin \alpha = \rho_\infty U_\infty^2 A \sin^3 \alpha \Rightarrow C_d = 2 \sin^3 \alpha \Rightarrow \frac{L}{D} = \cot \alpha$$

non linear even for small α



1. C_ℓ increases nonlinearly reaches max at $\alpha \cong 54.7^\circ$ then decreases. This is realistic for hypersonic flows

Remember: for low speeds $\alpha \cong 14^\circ - 16^\circ$ for $C_{\ell \text{ max}}$

2. C_ℓ varies nonlinearly with the angle of attack $\Rightarrow$ hypersonic aerodynamics is a nonlinear phenomenon
3. $L/D \Rightarrow \infty$ as $\alpha \rightarrow 0$ but when skin friction is added to the picture D becomes finite at $\alpha = 0 \Rightarrow L/D$ reaches a max at some small angle α and then decreases to zero at $\alpha = 0$
(---- line for $Re = 3 \times 10^6$, $M = 20$) This max ~ 8 is much smaller than subsonic values

Find $C_{\ell_{\max}}$

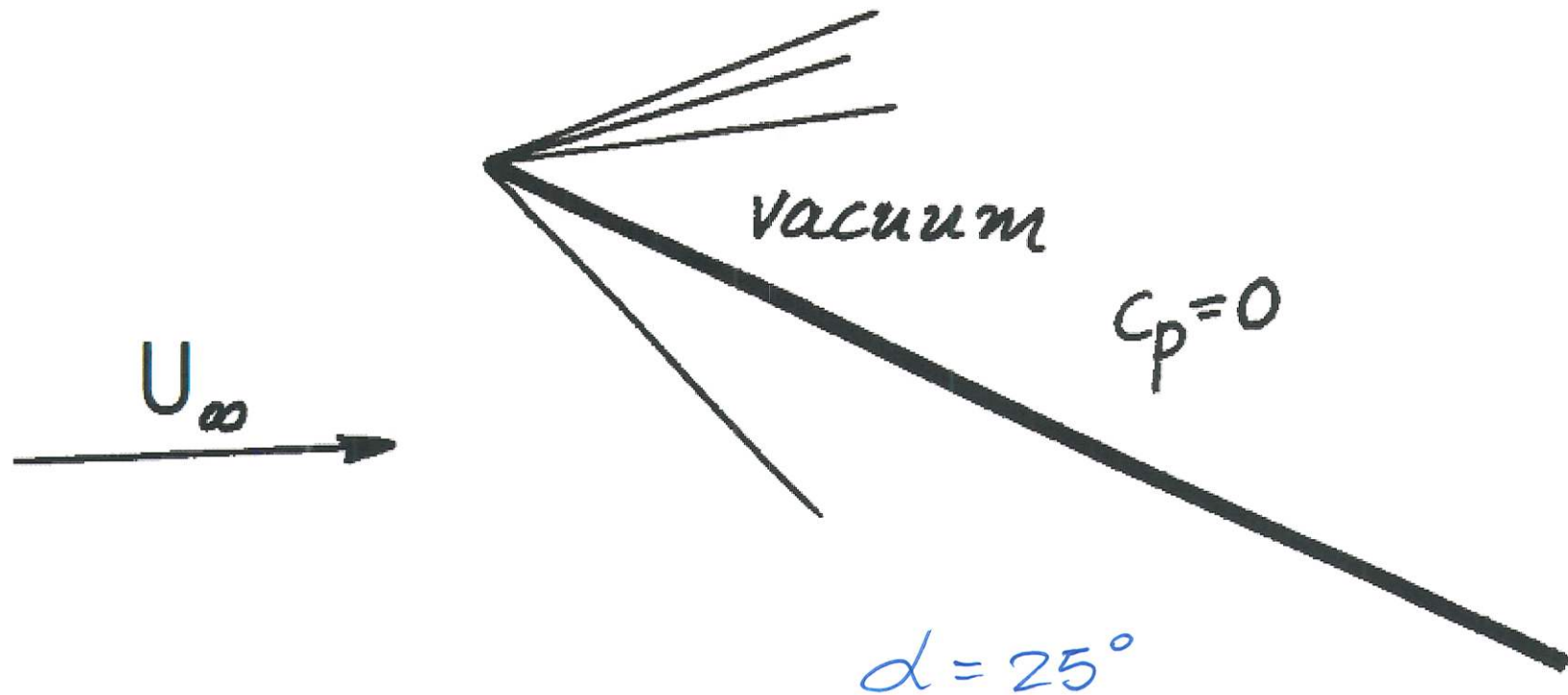
$$C_{\ell} = 2 \sin^2 \alpha \cos \alpha \quad \Rightarrow \quad \frac{dC_{\ell}}{d\alpha} = 2 \sin^2 \alpha (-\sin \alpha) + 4 \cos^2 \alpha \sin \alpha = 0$$

$$\Rightarrow \sin^2 \alpha = 2 \cos^2 \alpha = 2(1 - \sin^2 \alpha) \Rightarrow \sin^2 \alpha = \frac{2}{3} \Rightarrow \alpha = 54.7^\circ$$

$$C_{\ell_{\max}} = 2 \sin^2(54.7) \cos(54.7) = 0.77$$

Example: Flat plate at $\alpha = 25^\circ$ $M = 25$ find C_{ℓ} , C_d using

- straight Newtonian theory
- exact oblique shock & exp. waves



a) Newtonian Theory: $C_\ell = 2 \sin^2 \alpha \cos \alpha = 0.323$

$$C_d = 2 \sin^3 \alpha = 0.151$$

b) Exact: On the top surface – expansion fan

$$\nu_2 = \nu_1 + \theta = 119 + 25 = 144^\circ$$

from tables ($M=25$)

beyond max expansion angle $\Rightarrow$ “vacuum” $\Rightarrow C_p = 0$

on the top surface

bottom surface: oblique shock $\delta - \theta - M$ diagram $\Rightarrow \theta = 30.86^\circ$

$$(8.79) \quad C_p = \frac{4}{\gamma+1} \left(\sin^2 \theta - \frac{1}{M_1^2} \right) = 0.436$$

(note: $\theta = 1.28$ for high M_∞ ($M_\infty \rightarrow \infty$))
 $M \rightarrow \infty$
 δ, θ small (8.73) $\left(\theta \rightarrow \frac{\delta+1}{2} \delta \right)$

$$C_n = \frac{1}{c} \int_0^c (C_{p_l} - C_{p_u}) dx = 0.22 \Rightarrow C_l = C_n \cos \alpha = 0.395; C_d = C_n \sin \alpha = 0.185$$

Newton's law off:

22.3

13.9% for C_l

21.9

13.9% for C_d

$\theta = 20.28^\circ$

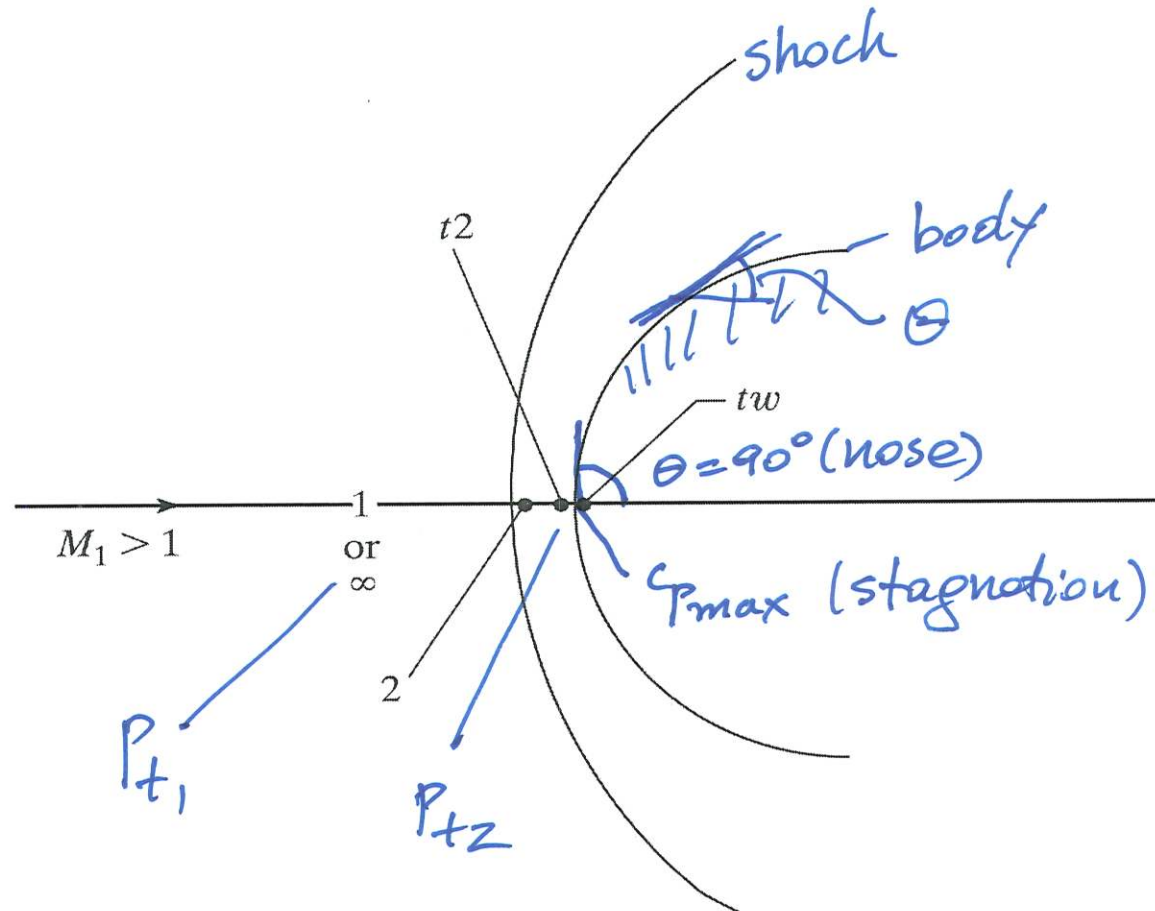
from 8.73 $\left(\begin{array}{l} M_1 \rightarrow \infty \\ \gamma \rightarrow 1 \end{array} \right)$
 $\theta \rightarrow \alpha$

$$C_p = 2 \sin^2 \theta$$

($\sim$ redo oblique shock for $\gamma=1^+ \sim$)
 result would be close

Properties of the stagnation region [12.3]

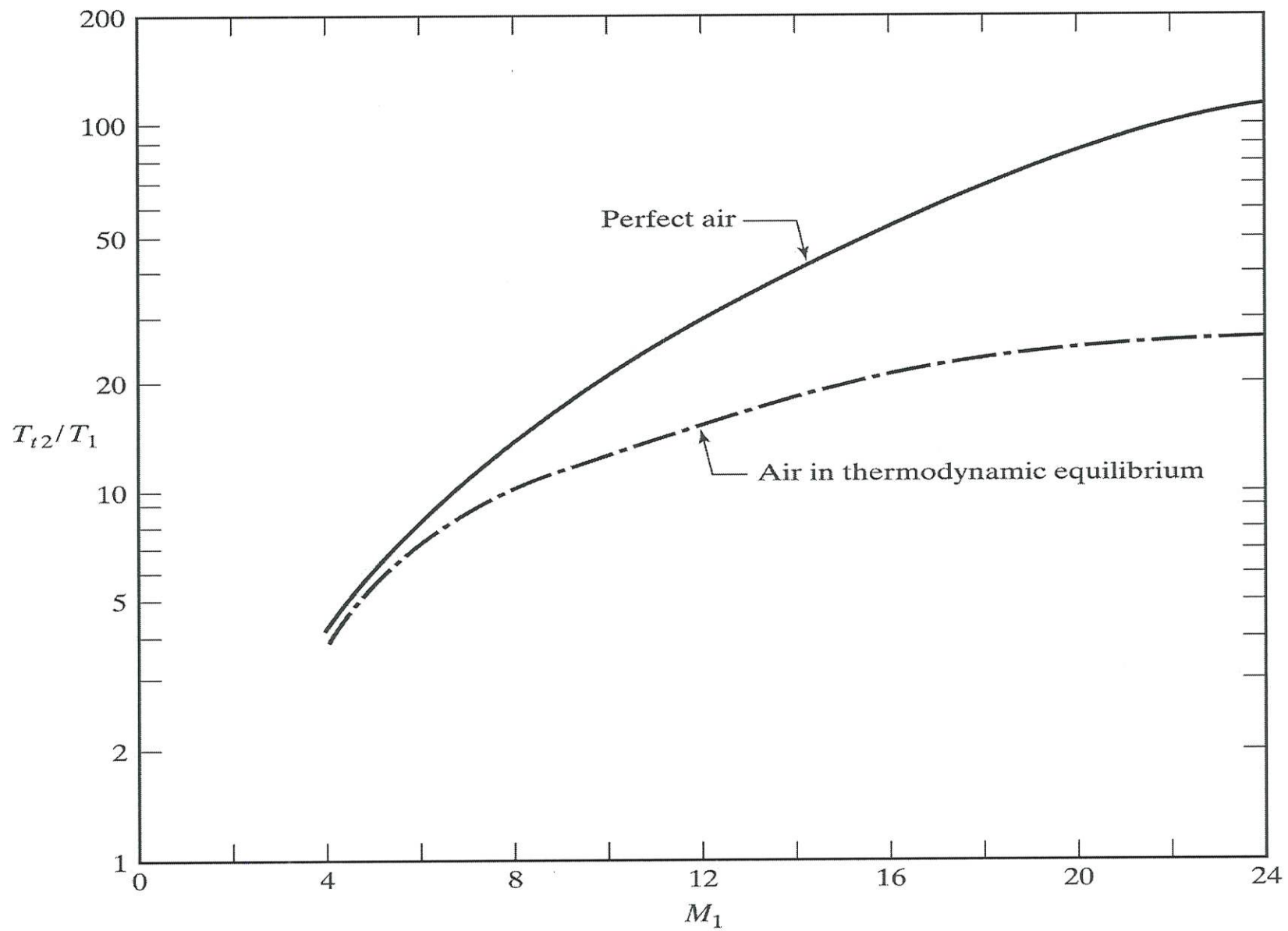
Normal shock relations (Chapter 8) still valid.



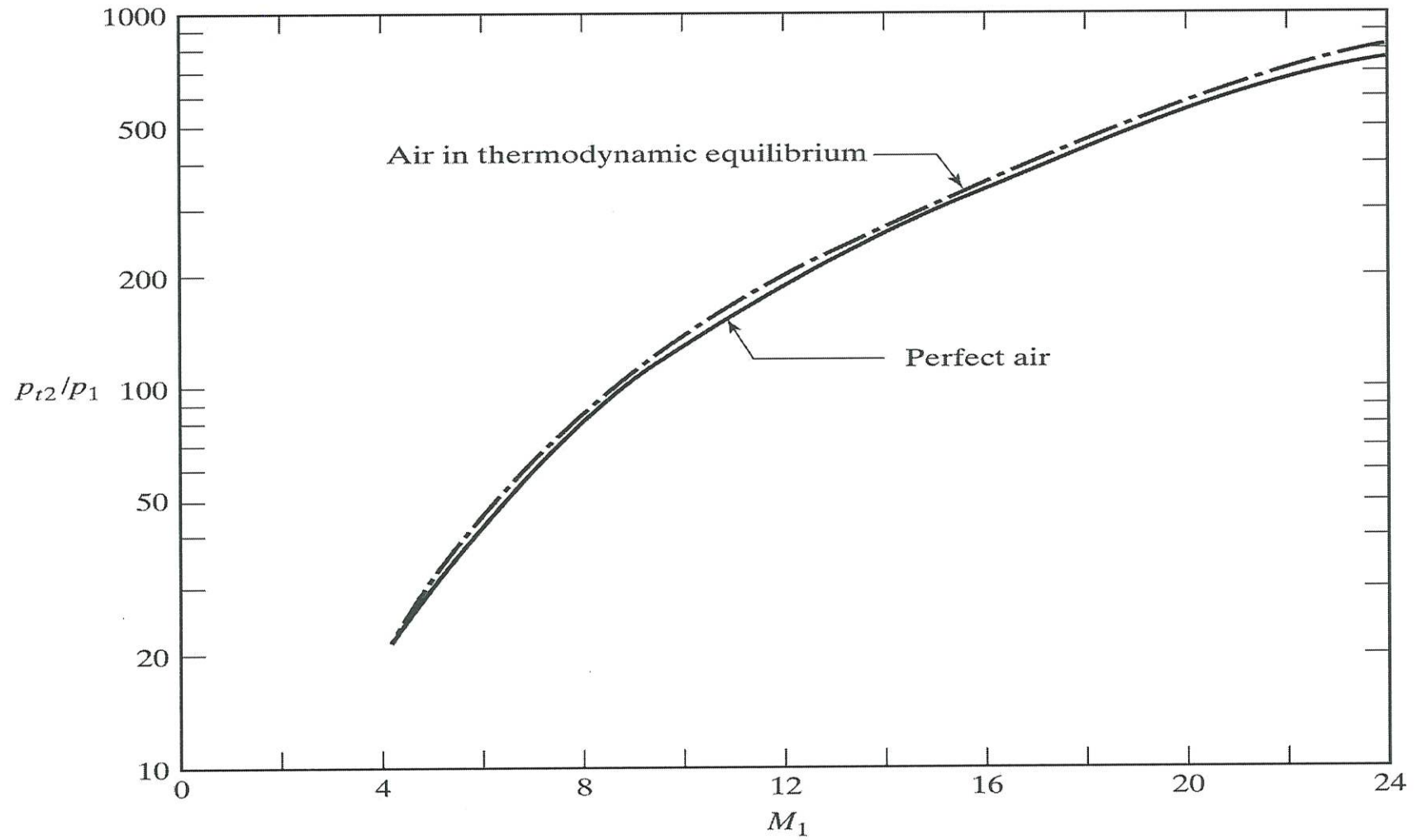
For higher temperatures (hypersonic flow) γ decreases
higher vibrational and chemical energy modes,
(non-equilibrium flow)

Results for different ratios

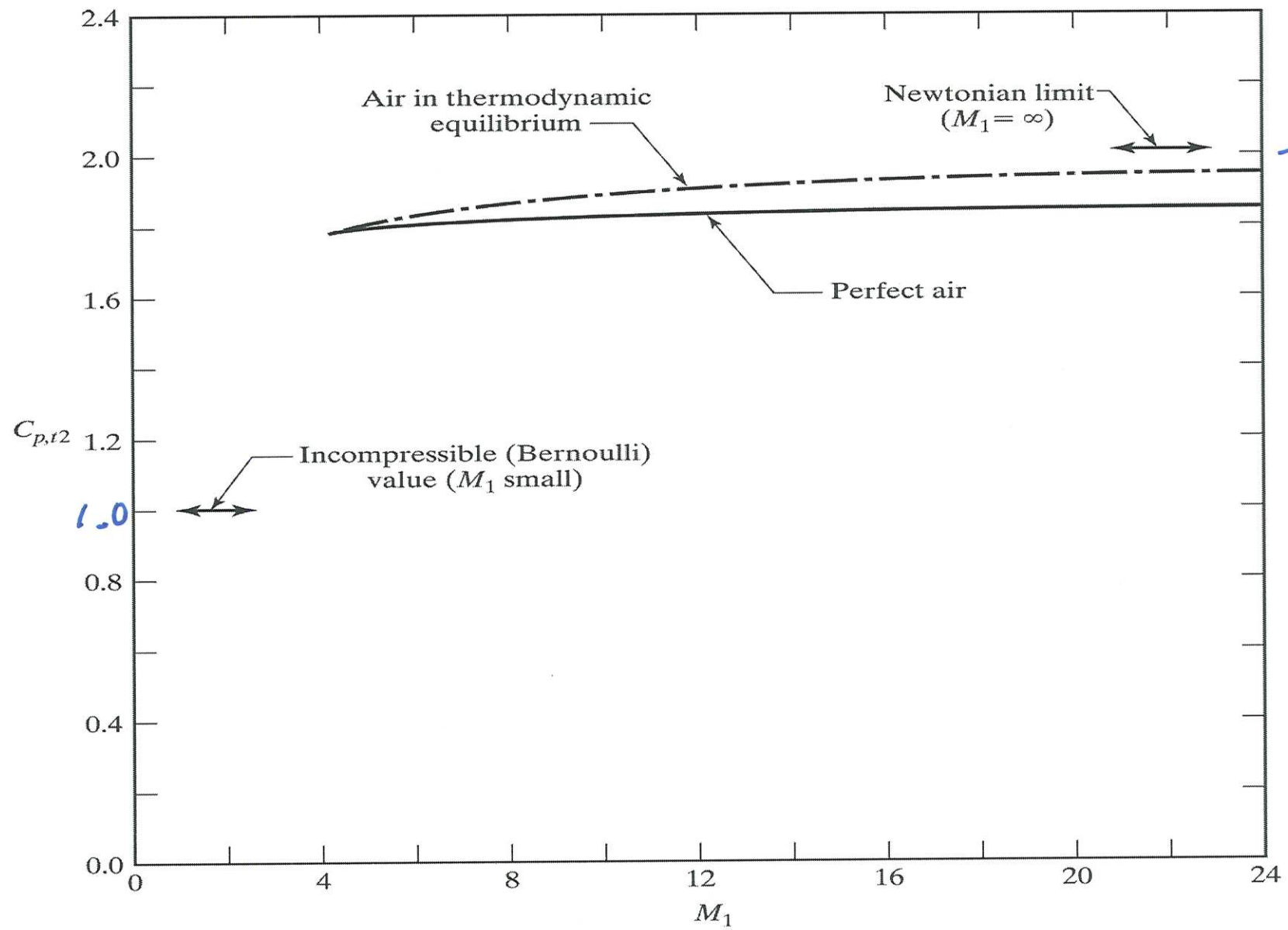
for 45,721 m (150,000 ft) for a sphere $R = 0.3048\text{m}$
at stagnation point



temperatures are different



pressures not that different



Modified Newtonian Theory [12.4]

2 can be replaced by C_p at stagnation point *(see also previous graph)*

$\Rightarrow$ modified Newtonian law:

$$C_p = C_{p,t2} \sin^2 \theta_b = C_{p,t2} \cos^2 \phi \quad (12.17)$$

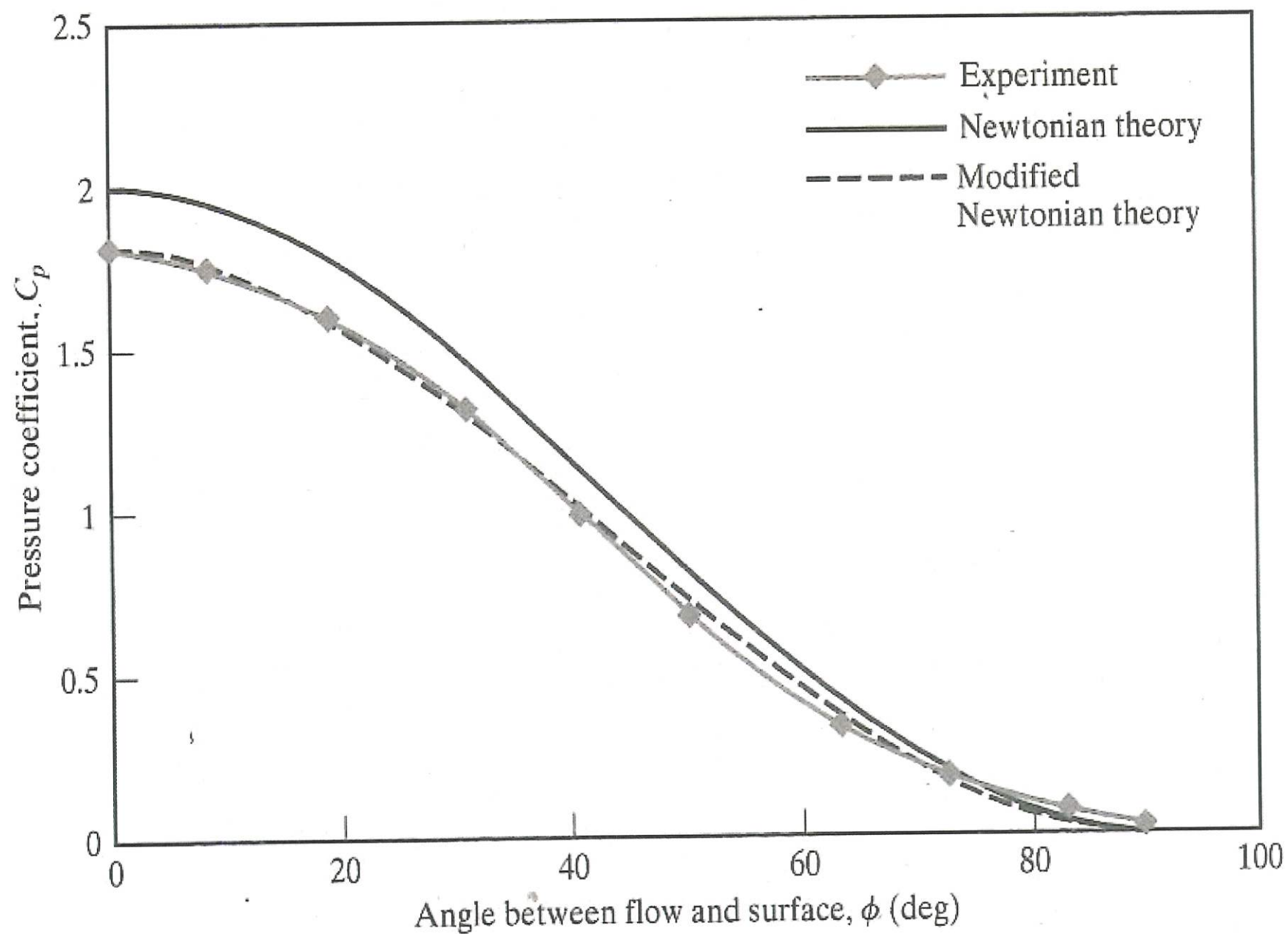
(approximates better conditions near the nose $\theta = 90^\circ$)

$$C_{p,t2} = \frac{p_{t2} - p_1}{\frac{1}{2} \rho_1 U_1^2} \Rightarrow C_{p,t2} = \frac{2}{\gamma M_1^2} \left(\frac{p_{t2}}{p_1} - 1 \right)$$

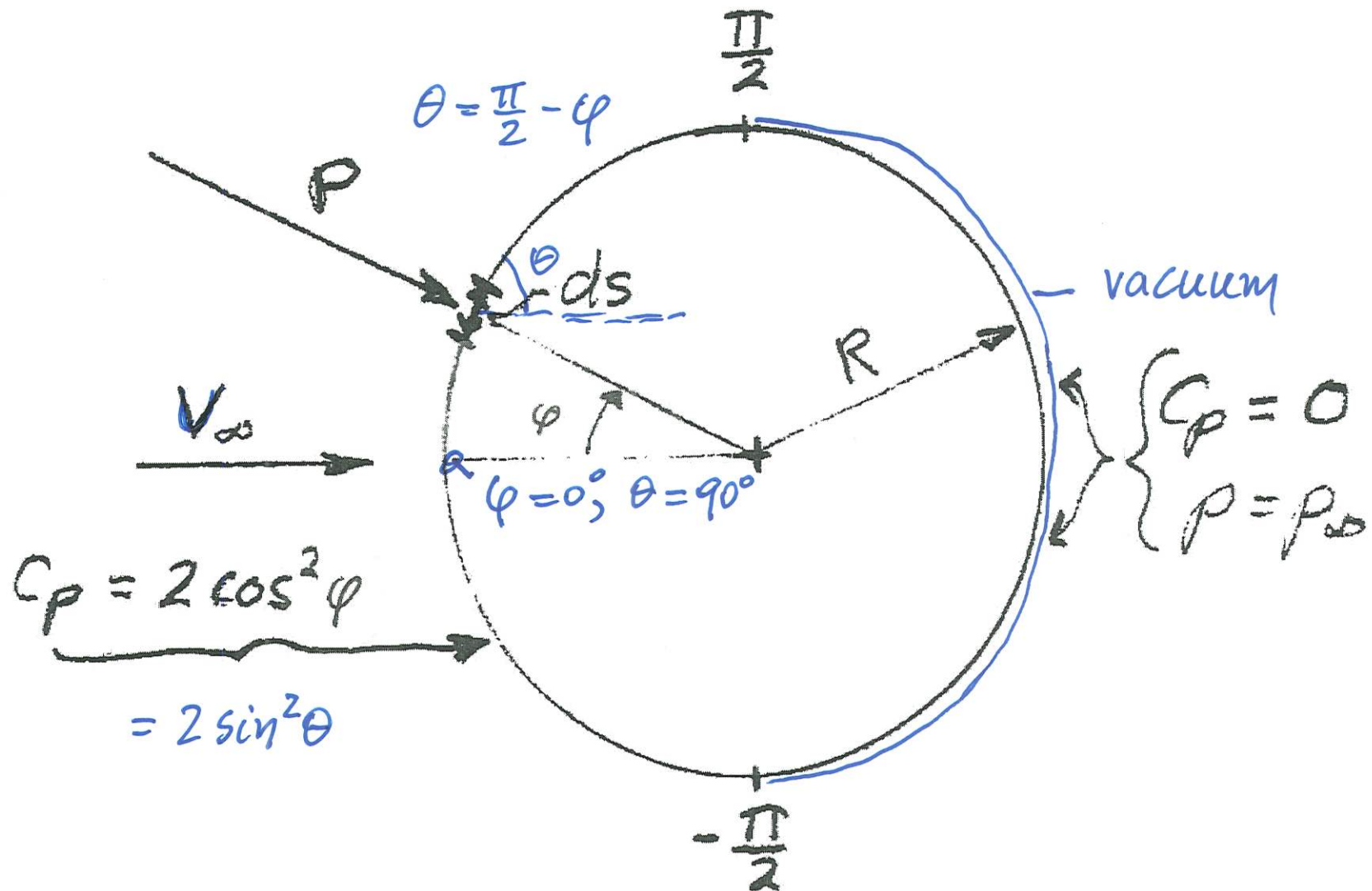
refer to figure in p. 29

(- for incompressible $C_p(\text{stagn.}) = 1$)

(12.17) compares well with CFD and experiments



Evaluate the C_d of a cylinder using Newtonian theory:



$$\int_0^{2\pi} p_\infty \cos \phi R d\phi = p_\infty R \int_0^{2\pi} \cos \phi d\phi = p_\infty R [\sin \phi]_0^{2\pi} = 0$$

$$D' = \int_0^{2\pi} (\underbrace{q_\infty C_p + \overset{0}{p_\infty}}_{\substack{\downarrow \\ p}}) \cos \phi R d\phi$$

($\rightarrow$ to get horizontal force $= \sin \theta$)

$$D' = q_\infty \int_0^{2\pi} C_p \cos \phi R d\phi \Rightarrow$$

$$C_d = \frac{D'}{q_\infty (2R)}$$

- only the front

$$\Rightarrow C_d = \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} C_p \cos \phi d\phi$$

" $2 \cos^2 \phi$ "

$$\Rightarrow C_d = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 \phi R d\phi$$

" $\sin^3 \theta$ "

$$= \frac{1}{3} [\sin \phi (\cos^2 \phi + 2)]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{3} [2 - (-1)2] = \frac{4}{3}$$

$$\Rightarrow C_d = \frac{4}{3}$$

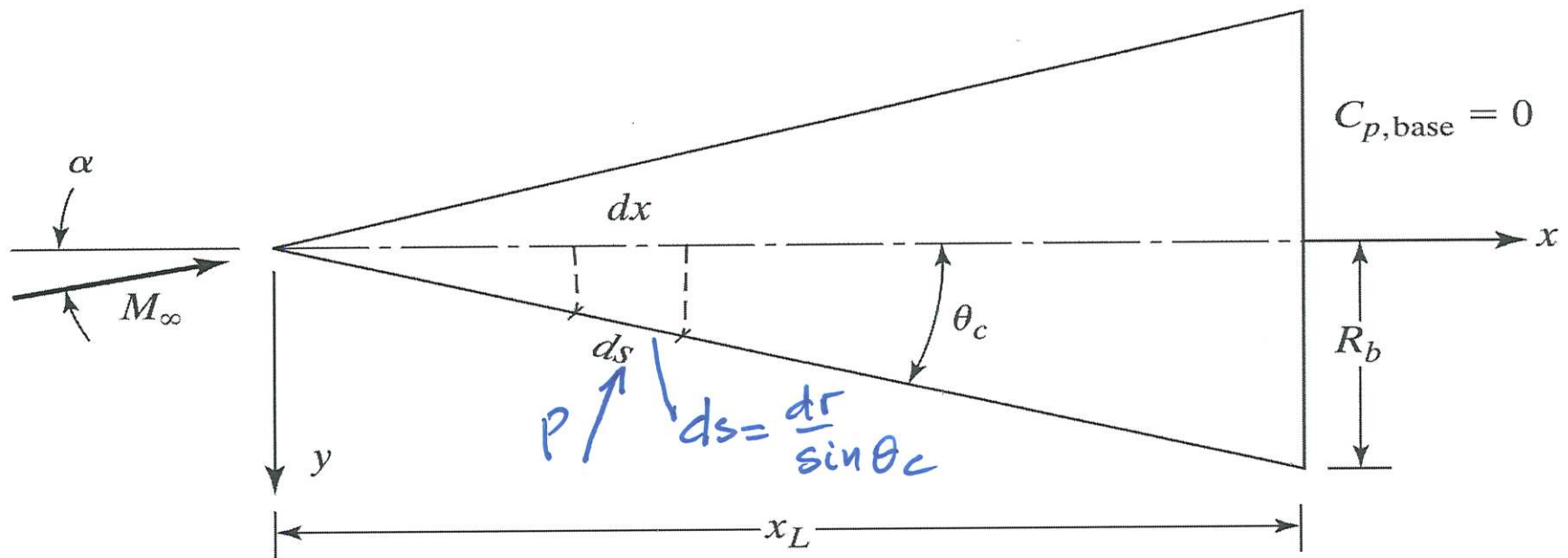
for modified Newtonian:

$$C_d = \frac{2}{3} C_{pmax}$$

which one do you think is more accurate? - modified better for blunt bodies

For a sphere, see text $C_d = 1$ - straight better for sharp-nosed bodies

$$= \frac{1}{2} C_{pmax}$$



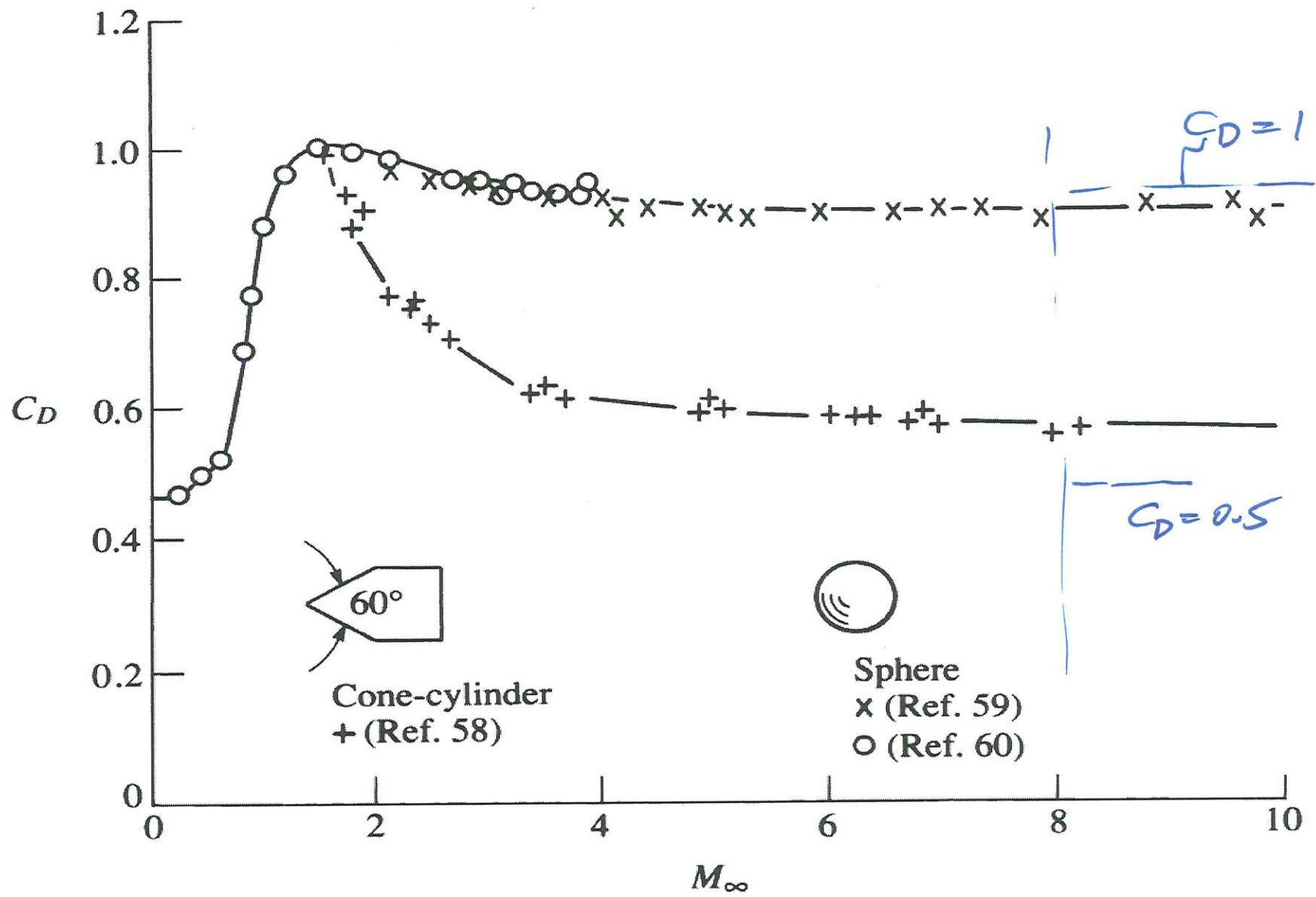
Cone at $\alpha=0$: Drag force on the strip: $(p \sin \theta_c)(2\pi r) \frac{dr}{\sin \theta_c} = 2\pi r p dr$
back surface *to get horizontal*

$$D = \int_0^{r_b} 2\pi r p dr - \int_0^{r_b} 2\pi r p_\infty dr = \pi (p - p_\infty) r_b^2$$

$$C_d = \frac{D}{q_\infty \pi r_b^2} = \frac{\pi (p - p_\infty) r_b^2}{q_\infty \pi r_b^2} = C_p$$

$$C_p = 2 \sin^2 \theta_c$$

for Newtonian



Newtonian theory: $C_p = 2 \sin^2 \theta$ (independent of M_∞)

Cone: $C_D = C_p = 2 \sin^2 \theta$ for $\theta = 15^\circ$ $C_p = 0.134$ [p. 21]

Cone-cylinder: $\theta = 30^\circ \Rightarrow C_D = 0.5$ [p. 40]

Sphere: $C_D = 1$

Both sphere & cylinder show Mach # independence.

Dimensional variables are not M number independent

Hypersonic waveriders (12.5)

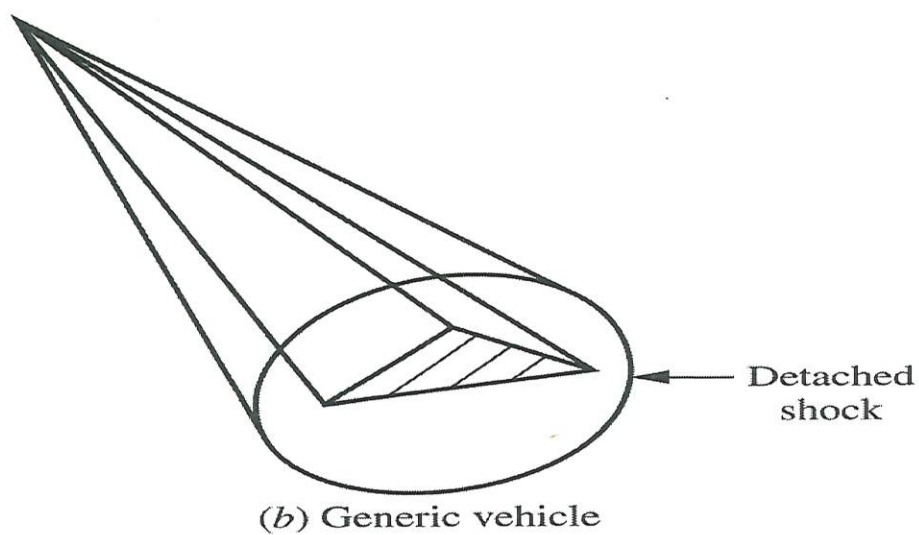
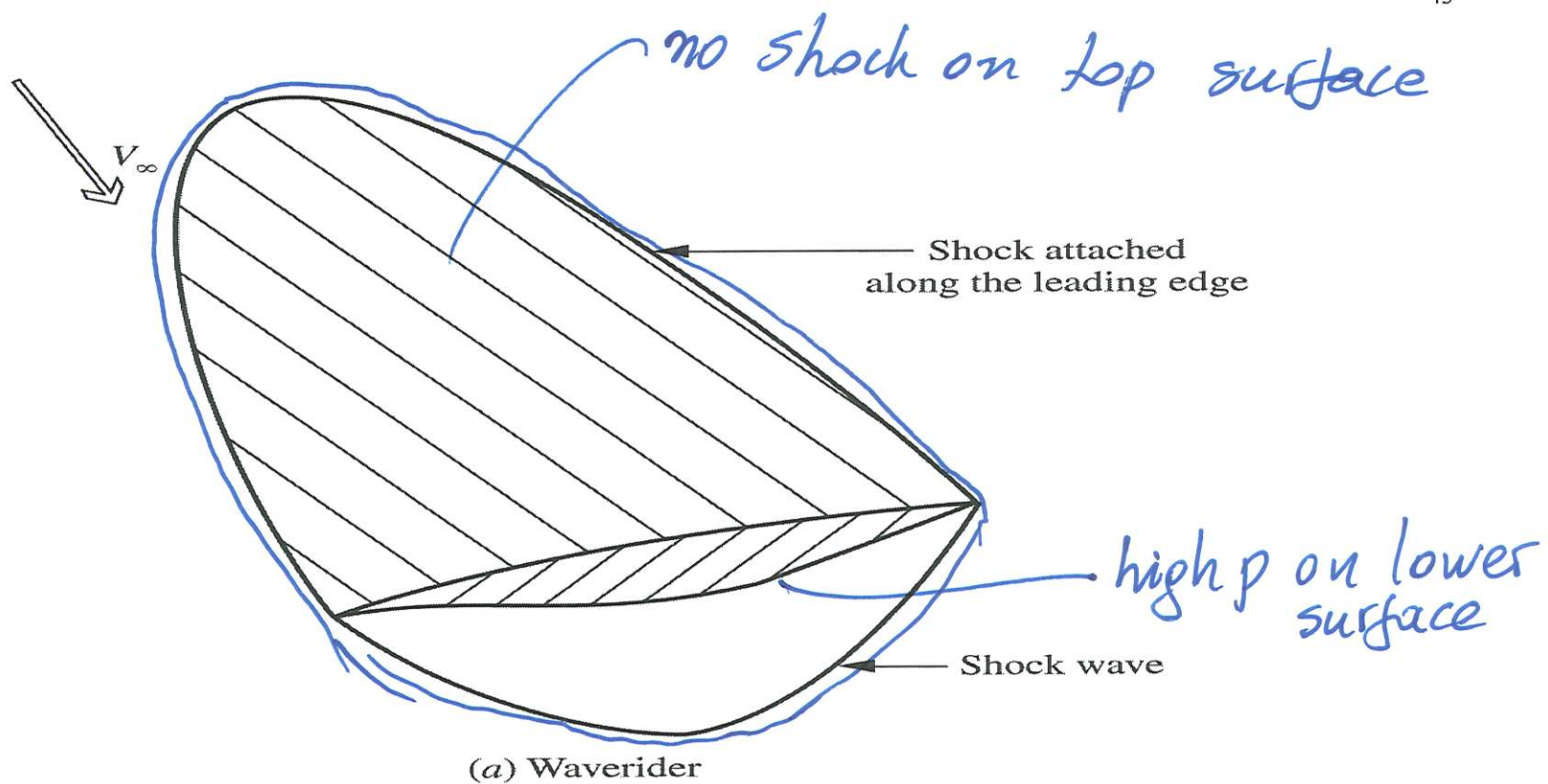
L/D value for hypersonic vehicles is low; can be increased using waveriders, i.e. shock attached along leading edge; vehicle is riding on the top of its shock wave.

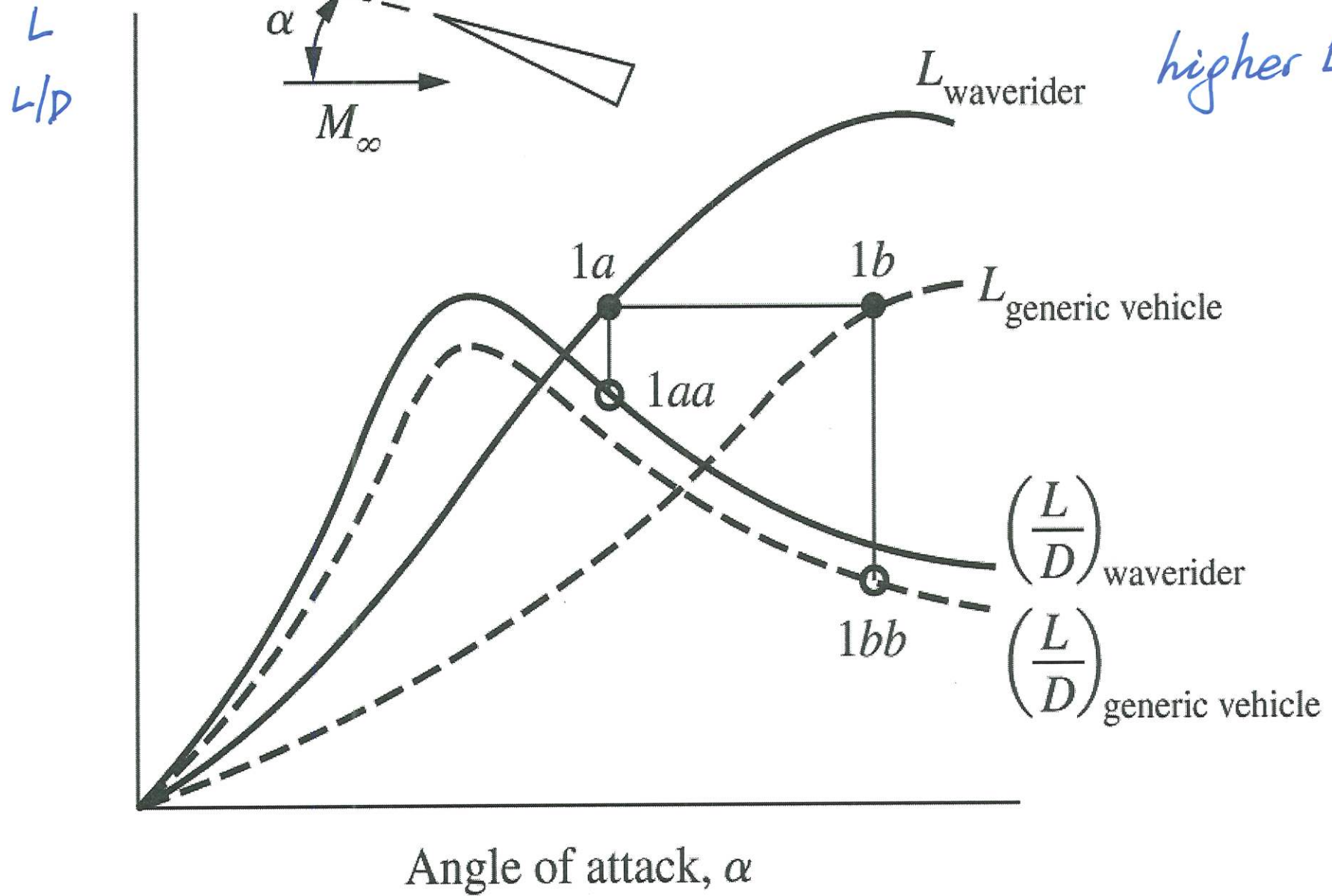
No leakage of high pressure, higher L/D *compared to a generic vehicle*
 wedges, cones used as base-flows (top surface parallel to the free stream, lower surface has attached shock) *→ higher pressure behind the shock → higher L*

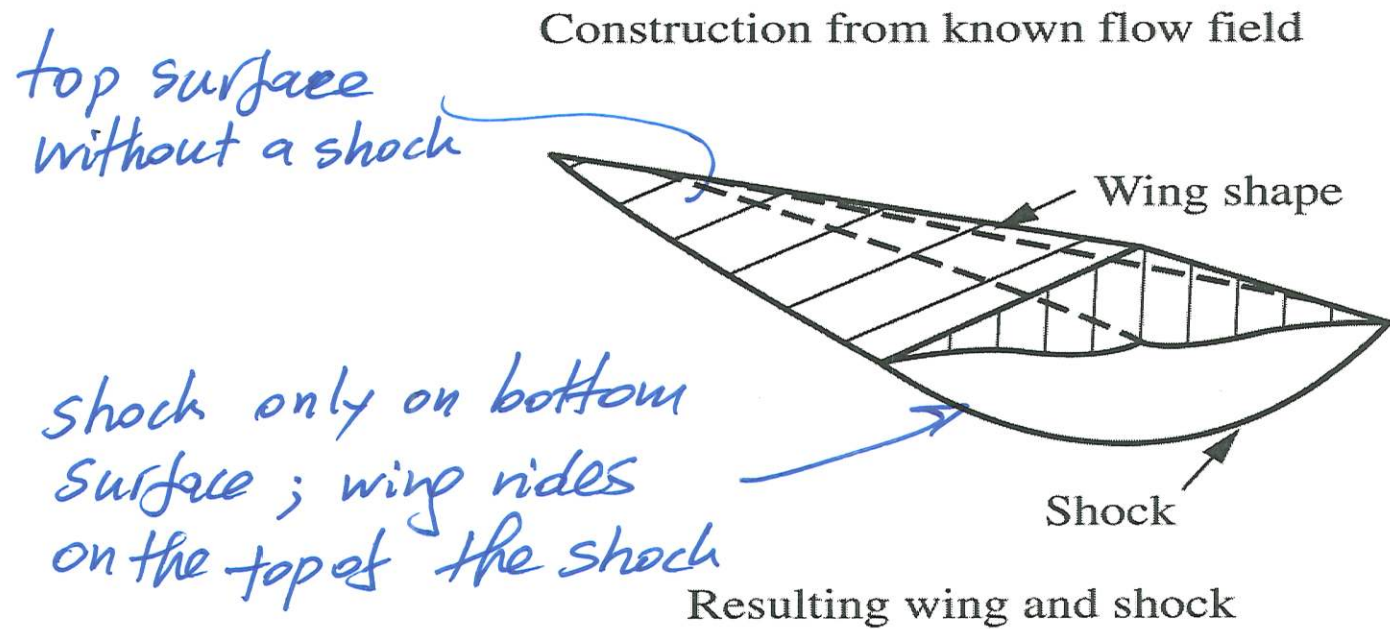
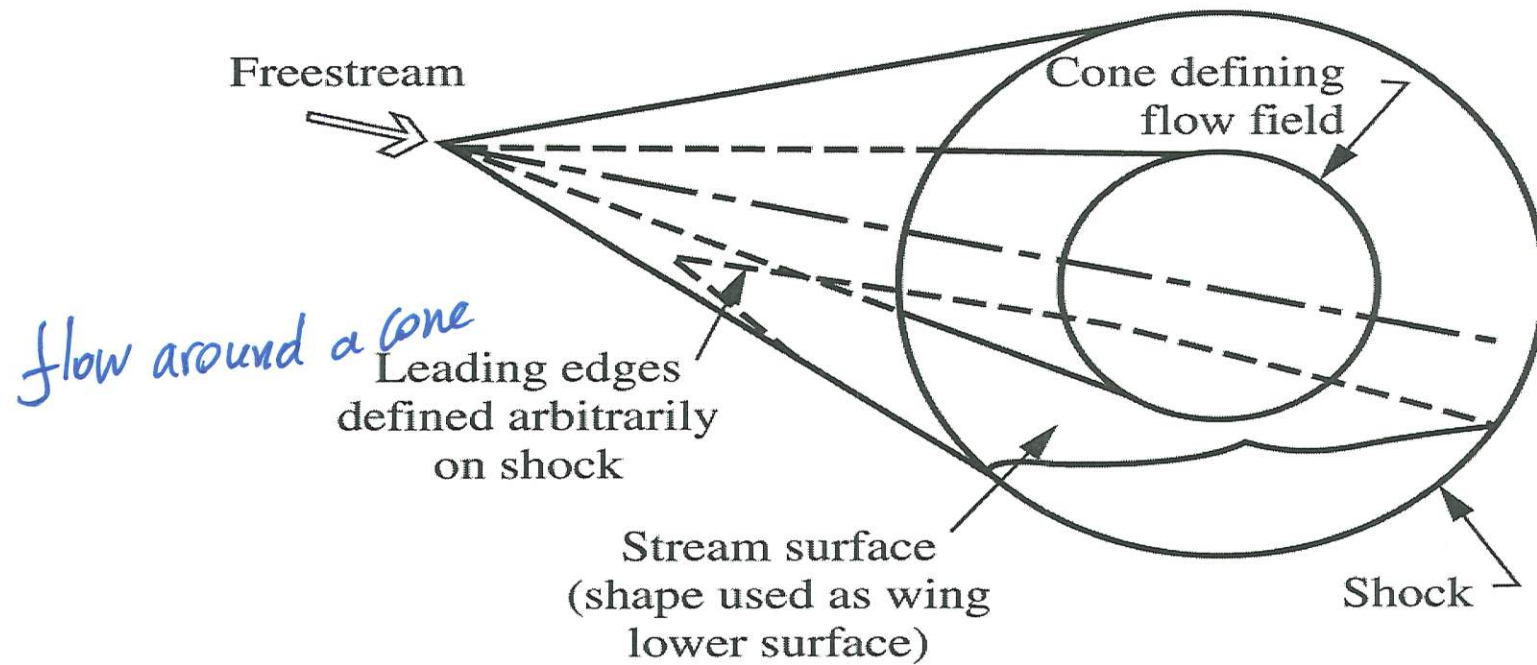
Shapes can be optimized adding viscous flow effects (e.g. X51 waverider) – active area of research

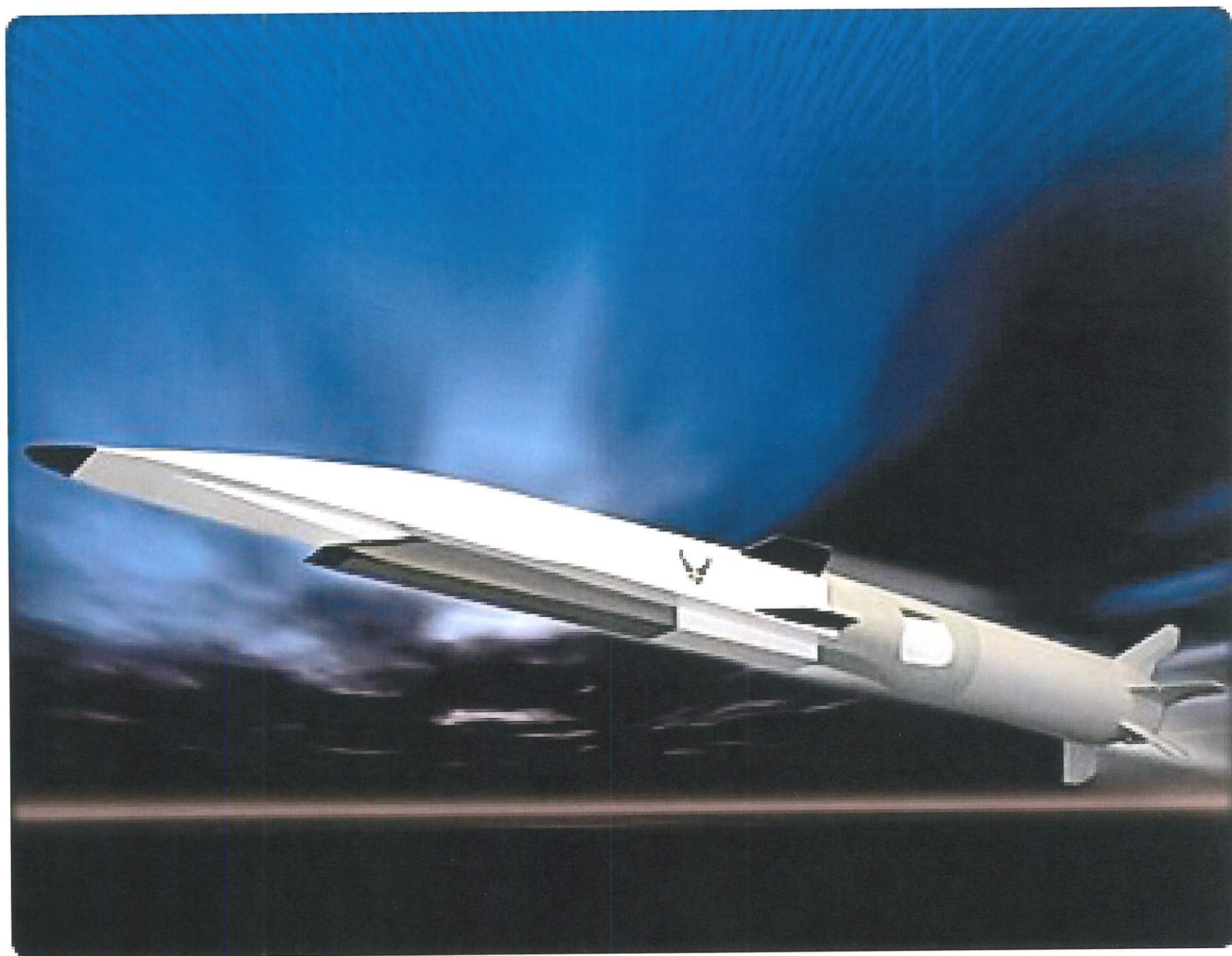
Note: design for a certain value of speed (or M) and α

(one point design)









Aerodynamic Heating (12.6)

Modes of Heat Transfer

conductive : Heat transfer by random molecular motion

Convective : Heat transfer by organized motion of the fluid
(due to buoyancy)

Radiative : Heat transfer by electromagnetic radiation

For Apollo re-entry: $\dot{q}_{\text{rad}}$ is 30% of the total $\dot{q}$

Convective Heat Transfer

$\dot{q}_w$ = wall heat flux *(rate of heat transfer/unit area)*

$$= k \left(\frac{\partial T}{\partial y} \right)_w \quad \text{Fourier's law, } k = \text{thermal conductivity}$$

Nondimensionalize:

$$C_H = \frac{\dot{q}_w}{\rho_{\infty} V_{\infty} (h_{aw} - h_w)} = \text{Stanton number}$$

*non-dimensional #
for heat transfer*

h_w = *wall enthalpy*

h_{aw} = *adiabatic wall enthalpy (insulated wall)*

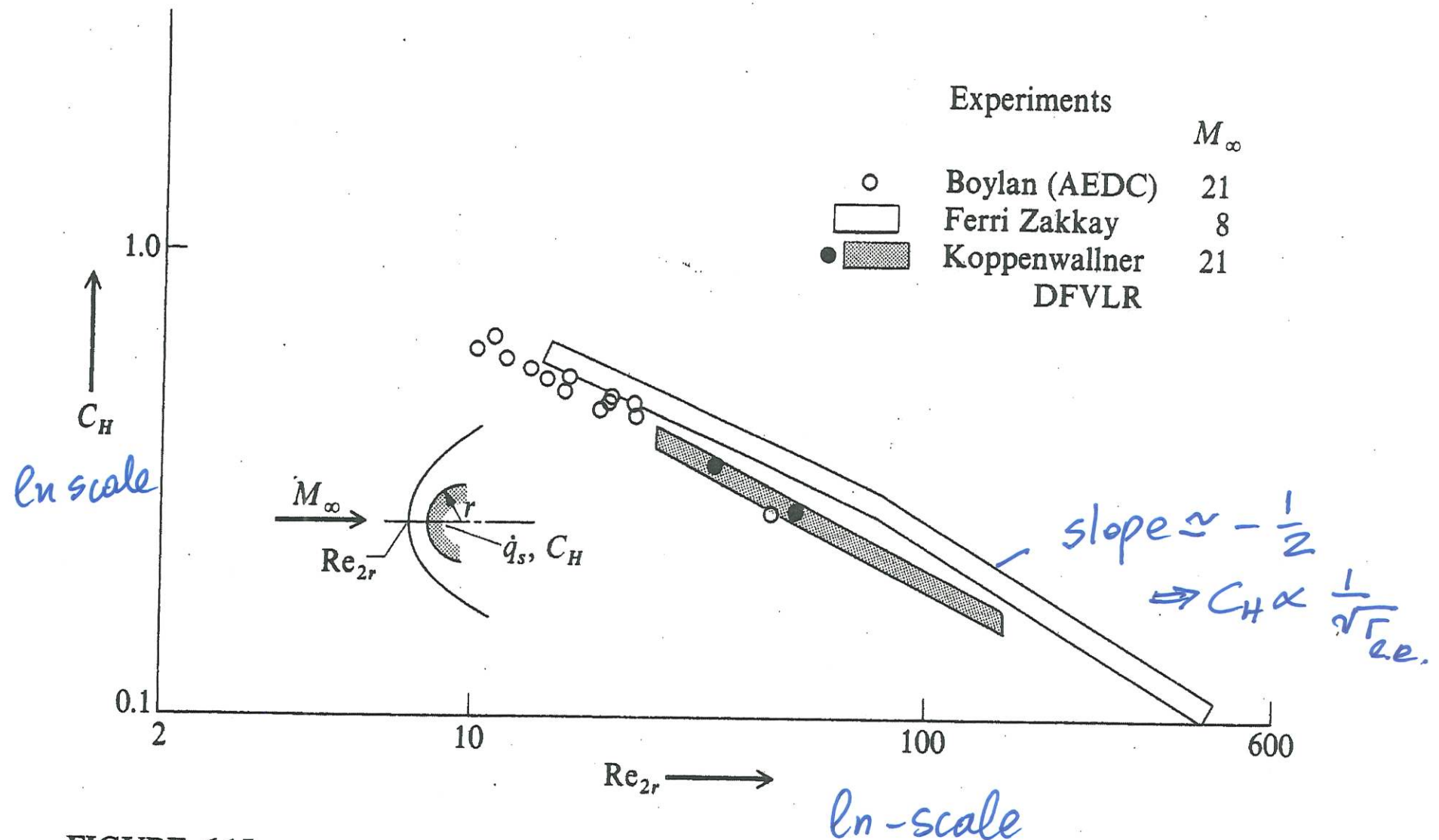


FIGURE 6.15

Stagnation point Stanton number versus Re based on nose radius. (From Koppenwallner, Ref. 79.)

Boundary Layer Transition (12.9)

Transition influences the heat rate (see next figure)

Estimates: 20-80%, affecting the gross weight by a factor of > 2

